

Drosophila ellipsoid body ring neurons instruct opposing
internal states that remodel physiology and slow aging

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9 Abstract

10 Early studies of aging processes emphasized cell-autonomous mechanisms, but it is now well-established
 11 that some aspects of lifespan modulation are neurally mediated. Little is known, however, about the
 12 nature of the brain's involvement in aging or the mechanisms underlying these contributions. Un-
 13 derstanding the neural regulation of aging has important implications for health, as it represents a
 14 theoretical upstream point of intervention in multiple aging processes. A neural locus shown to influ-
 15 ence aging in multiple contexts would offer a promising proof of this concept. Our lab has identified
 16 one such locus in the ellipsoid body (EB), a neuropil in the *Drosophila* brain. The EB, whose diverse
 17 functions include behavioral regulation, sensory integration, and memory formation, is a component
 18 of the central complex (CX), a region functionally and structurally analogous to the mammalian basal
 19 ganglia (BG). We have found that manipulation of EB activity reverses the deleterious effects of mul-
 20 tiple stressful environmental conditions and increases lifespan. Our data indicate that ER2 and ER4
 21 ring neurons in the EB are particularly influential in modulating these phenotypes. We have discov-
 22 ered that direct manipulation of ER2 and ER4d neurons is sufficient to extend lifespan, potentially
 23 via insulin-like signaling and other known aging pathways. These manipulations establish distinct
 24 metabolic and transcriptional profiles and are experienced with opposing valence, suggesting that dif-
 25 ferential circuit recruitment establishes distinct internal states. Whether these states engage distinct
 26 aging-related pathways or converge on shared peripheral mechanisms remains under investigation. We
 27 expect that a deeper understanding of how this neural integration center interfaces with conserved
 28 pathways will reveal broadly applicable principles underlying neural control of aging. Given the func-
 29 tional homology between the CX and mammalian BG, we expect these findings to provide a starting
 30 point for identifying similar mechanisms in mammalian brains.

31 Introduction

32 Over the past several decades, research across diverse model organisms, from yeast and nematodes to
 33 flies and mammals, has revealed that the rate of aging is not fixed and can be modulated by genetic,
 34 environmental, and pharmacological interventions [1]. Remarkably, many of the molecular pathways
 35 that regulate lifespan are highly conserved, suggesting that fundamental mechanisms of aging may

be shared across taxa [2]. While much of aging research has involved direct perturbation of these conserved pathways, an intriguing dimension of aging biology was revealed by the discovery that sensory perception can modulate lifespan even in the absence of the behavioral interactions typically associated with the perceived stimulus [3, 4, 5, 6]. Studies in the nematode worm, *Caenorhabditis elegans*, and the vinegar fly, *Drosophila melanogaster*, have demonstrated that sensory cues related to food, mates, and even the presence of dead conspecifics can initiate physiological changes that influence rates of aging [7, 8, 9, 10, 11]. These effects are mediated by neuronal circuits that emanate from sensory tissues and interface with deeper regions of the central nervous system, ultimately engaging conserved neuromodulators, including biogenic amines and neuropeptides, that influence metabolism, stress resistance, and longevity [8, 5, 11].

In certain instances, sensory perception has been shown to modulate lifespan through the internal and motivational states that it influences. For example, the neuropeptide F system, homologous to mammalian neuropeptide Y, is a component of reward- and stress-related processing, and its activation can reduce starvation resistance and shorten lifespan [12, 7]. Sensory perception of potential mates shortens lifespan in male flies through circuits involving NPF signaling, whereas successful mating reverses these effects [7], suggesting that an unresolved expectation of reward can influence lifespan. Insatiable hunger also slows aging, despite promoting behaviors normally associated with decreased lifespan [13]. These observations indicate that identifying the neural circuits that interpret and integrate sensory inputs, as well as how they generate conserved internal states that direct physiological outcomes, may provide new insight into healthy aging.

In *Drosophila*, one neural substrate has recently emerged as a strong candidate for such an interpretive center: the ellipsoid body (EB). The EB resides within the central complex, a region that shares functional and organizational homology with the mammalian basal ganglia [14]. The EB is subdivided into concentric domains innervated by several classes of ring neurons (termed "ER" neurons) that process visual, thermal, mechanosensory, and circadian information to drive behavioral outputs [15, 16, 17]. Additional roles for EB ring neuron classes involve locomotor control, sleep, visual memory, and state-dependent action selection, suggesting that it is well-positioned to link myriad sensory representations to organismal state [18, 17, 16, 19]. Recent work has implicated this structure in aging biology. Exposure of flies to dead conspecifics, termed "death perception," reduces fat stores,

lowers starvation resistance, and shortens lifespan in a manner that requires visual function and serotonin signaling through 5-HT_{2A}-expressing ER2 and ER4 neurons [8]. Insulin-like peptides, as well as the insulin-responsive transcription factor dFOXO, are required for aging effects of death perception, suggesting that EB ring neurons interface with conserved metabolic pathways implicated in lifespan control [20].

The extent to which the results in *Drosophila* may illuminate similar processes in other taxa may hinge on the nature of the EB's role in influencing health-related outcomes. One possibility is that these neurons function primarily as passive conduits or gates for lifespan-altering sensory information: if their function is disrupted, flies may simply fail to perceive [21, 22] or internally represent salient stimuli, and phenotypes normally induced by those stimuli would not manifest. A more intriguing alternative is that the EB is an instructive modulator of aging-related processes, and direct manipulation of ring neuron activity, independent of particular sensory input, would be sufficient to alter aging. If true, the EB and functionally similar structures—i.e., central hubs of neuromodulation and sensory integration—would serve as rich targets for aging research.

Here, we use a variety of optogenetic tools, which enable temporally precise activation or inhibition of genetically-defined neuronal populations, to show that *Drosophila* EB ring neuron populations are instructive modulators of aging that encode behaviorally meaningful information and orchestrate diverse transcriptional and metabolic responses throughout the animal. Activation of ER2 neurons, inhibition of ER4d neurons, and simultaneous activation of ER2 and ER4 neurons each extends lifespan in males and females, but the manipulations produce divergent transcriptional responses in head tissues, including distinct effects on insulin signaling, stress-response, and metabolic pathways. Despite these molecular differences, we detect no consistent changes in feeding, fat storage, or climbing ability that account for the longevity effects, arguing against a simple explanation based on overt energetic or behavioral shifts. Behavioral assays further show that ER2 activation alone produces a negatively valenced state, whereas combined ER2/ER4 activation produces a positively valenced state, indicating that distinct neural states with different immediate behavioral consequences can converge on increased lifespan. Transcriptomic and metabolomic profiling of peripheral tissues further reveals that ER2 activation drives systemic metabolic remodeling, including suppression of proliferative programs and upregulation of electrochemical and membrane homeostasis pathways, demonstrating that EB ring

neuron activity induces coordinated responses in the periphery. Together, these findings support a model in which the ellipsoid body serves as an integrative neural locus through which sensory and state-related processes act as active regulators of pro-longevity physiology via multiple downstream processes.

Results

Manipulating ER2 and ER4 neurons in the ellipsoid body extends fly lifespan

To determine if the activity of ellipsoid body ring neurons is capable of modulating lifespan, we performed a series of experiments in which the activity of EB neurons was manipulated, with the following considerations informing our experimental design. First, we sought to eliminate potential developmental effects and to restrict neuronal manipulations to adults with a developed central complex [23]. Second, we wanted to avoid tools that require introducing confounding factors, such as high temperatures, which can significantly affect lifespan and obscure treatment effects on aging [24]. Third, we wanted the flexibility to perform acute, cyclic, or chronic manipulations in ethologically-relevant contexts involving lifespan, behavior, and health assays. To accommodate these requirements, we adopted an optogenetic [25] approach to apply different treatment paradigms to freely-moving and fully developed adults by changing light-activation parameters and exposure times. To assay lifespans, we housed flies in custom optogenetic rigs with programmable, interchangeable LED bars, accommodating different channelrhodopsins and allowing us to either activate [26] or inhibit [27] targeted neuron populations. Unless otherwise noted, flies in all treatment paradigms were exposed to light activation (40 Hz, 32% duty cycle) during ZT 0-12 of a 12:12 light:dark cycle, beginning at 3 days post-eclosion and continuing throughout adulthood, promoting a normal circadian rhythm and restricting neuronal manipulations to periods when flies would typically be alert and interacting with their environment (Fig. 1A). Flies were otherwise maintained using standard husbandry conditions [28].

We focused our investigation on ER2 and ER4 neurons, with particular emphasis on the ER4d subpopulation, as these are required for the perception of dead conspecifics to modulate lifespan [20] and for food perception to enhance starvation resistance [29]. When flies expressing UAS-CsChrimson [26]

in ER2 neurons (ER2>Chr) were exposed to red light during subjective day, we observed a significantly extended lifespan in both females and males compared to siblings aged in the dark (Fig. 1B). This effect was not observed in genotype control flies that did not express CsChrimson, ruling out general lifespan-extending effects of red light and indicating a direct effect of neuronal activation. Activation of ER4d neurons had a modest but significant effect on lifespan (Fig. S1A). By contrast, inhibition of ER4d neurons by exposing flies expressing the anion channelrhodopsin GtACR1 [27] (ER4d>GtACR1) to green light during subjective day produced a more pronounced lifespan extension in both females and males (Fig. 1C). These results should be interpreted with caution, however, because we observed that exposure to high-intensity green light caused a significant reduction in lifespan in all cohorts (roughly 50% in females and 33% in males). Nevertheless, GtACR1 flies lived modestly but significantly longer than control flies in the light, a pattern that was not observed in the dark.

The observation that activation of ER2 neurons and, to a lesser extent, ER4d neurons increased lifespan was somewhat unexpected, given that measures of ER2 and ER4 neuronal activity were both observed to be higher following exposure to dead conspecifics [20], which accelerates aging [8]. We therefore decided to examine the effect of simultaneous activation of ER2 and ER4 (comprising ER4d and ER4m subpopulations) using a split-GAL4 driver. To our surprise, simultaneously activating both populations extended lifespan in males and females (Fig. 1D) to a greater extent than did ER2 activation alone. In this experiment, red light exposure modestly increased lifespan of CsChrimson-expressing flies as well as their genotypic controls, but the increase in lifespan was significantly greater in CsChrimson flies. Given the substantial increase in lifespan observed following simultaneous ER2/ER4 activation, we asked whether even broader manipulation of ellipsoid body neurons would recapitulate or even potentiate the lifespan extension that we observed following ER2 and ER2/ER4 neuron activation. Using a driver that targets additional EB neuron populations, including at least R3 neurons, we observed that broader EB inhibition had a modest effect on lifespan (Fig. 1E, left) and activation increased lifespan to an equal or lesser extent than ER2 or ER2/ER4 activation alone (Fig. 1E, right).

Together, these data demonstrate that manipulation of ER2 and ER4 neural activity increases lifespan, supporting the notion that the ellipsoid body is an instructive modulator of aging. These results may be interpreted in the context of known activity patterns of ER2/ER4 in other contexts.

Ishimoto et al. reported that ER2, ER4m, and ER4d neurons work together in an incoherent type 1 feedforward loop (I1-FFL) [30]. Upon simultaneous activation of ER2/ER4m and ER4d, ER4d signaling initially increased, but sustained activity of GABAergic ER2 and ER4m neurons gradually suppressed ER4d activity. If optogenetic activation recruits ER2/ER4 in this feedforward fashion, it may be that ER2 activation, ER2/ER4 activation, and ER4d inhibition are effectively the same intervention. Alternatively, they may engage partially or fully distinct pathways that converge on extended lifespan via different downstream mechanisms.

Different ring neuron manipulations produce distinct transcriptional profiles

To investigate how ER2 and ER4 neurons modulate aging and whether they do so via similar or distinct mechanisms, we performed bulk RNA-sequencing on brain-enriched tissues of female flies after one week of ER2 activation, ER4d inhibition, or ER2/ER4 activation, under the same conditions used in the lifespan assays described above. We chose to sample flies after one week of optogenetic manipulation to avoid survivorship bias that might arise at later time points while still allowing time for physiological differences to manifest. Flies were transferred directly from experimental rigs to liquid nitrogen, then decapitated by vortexing, and heads were collected with a sieve. Five biological replicates of 10 heads were collected for CsChrimson/GtACR1 and control genotypes from flies maintained either in constant darkness or channel-activating red/green lights during the 12 hours of subjective day. Whole head samples comprised primarily brain tissue, although other tissues were present, including the head fat body.

We used the DESeq2 package in R [31] to identify transcriptional changes associated with enhanced ER2 neural activity. We specified a statistical model in which genotype (CsChrimson, control) and light stimulation (yes, no) served as main effects. A genotype $\times$ light interaction term was used to identify significant transcriptional changes that were specific to light-driven channelrhodopsin stimulation, and these were the primary focus of our analysis. After one week of ER2 neuron activation, we identified 123 genes (70 downregulated and 53 upregulated) whose expression values exhibited experiment-wise interaction p-values (corrected for false discovery rate) less than our cutoff of $p < 0.01$. Expression data from these genes produced clear separation of samples by treatment, indicating strong reproducibility across each of the five replicates (Fig. 2A). Hierarchical clustering revealed that samples from flies with

178 activated ER2 neurons (i.e. ER2>Chr flies exposed to red light) formed a distinct group from flies
 179 without neuronal activation, including CsChrimson-expressing flies maintained in constant darkness
 180 as well as control genotype samples maintained in either darkness or red light.

181 Despite a stringent threshold, we observed that one week of ER2 neuron activation drove broad but
 182 pathway-enriched changes in the head. Gene Ontology (GO) analysis of upregulated genes revealed
 183 enrichment for ribosome biogenesis and rRNA processing, insulin signaling, and response to starvation
 184 (Fig. 2B). Downregulated genes were enriched for protein folding and ER stress response pathways
 185 (Fig. 2C). Together, these results suggest that activation of ER2 neurons induces a distinct metabolic
 186 state in the head, characterized by altered nutrient signaling and reduced ER stress. Notably, many
 187 of the differentially-expressed genes have been implicated in aging biology, with several known to
 188 modulate aging directly. For example, we observed the upregulation of multiple genes consistent with
 189 dampened insulin-like signaling (e.g., *foxo*, *Thor* (encoding d4E-BP), *InR*, *dilp6*) [32]; upregulation
 190 of autophagy-related genes such as *Atg101* [33]; upregulation of *ImpL2*, an IIS antagonist which has
 191 been linked to lifespan extension [34]; and higher levels of *Cbs*, which encodes the rate-limiting enzyme
 192 in the transsulfuration pathway and increases lifespan [35]. Together, these results demonstrate that
 193 short-term, part-time activation (12L/12D) of a few dozen ER2 neurons drives significant changes in
 194 multiple aging-, nutrient-, and stress-response pathways.

195 Following one week of ER4d silencing, far fewer changes in gene expression could be attributed to
 196 ER4d inhibition (Fig. 2D), although such changes were likely masked by remarkably potent indepen-
 197 dent effects of green light exposure. As observed for the ER2 data, sample-based clustering grouped
 198 biological replicates as more closely related to each other than to samples from all other treatments,
 199 demonstrating reproducibility. Consistent with its significant negative effect on lifespan (Fig. 1C),
 200 green light exposure affected the expression of 1,817 genes regardless of genotype (experiment-wise p
 201 < 0.01). However, only 11 genes (9 downregulated and 2 upregulated) exhibited a significant geno-
 202 type $\times$ light interaction, indicating a differential transcriptional response in flies with ER4d neurons
 203 silenced. Interestingly, in both genotypes, green light drove strong upregulation of multiple genes that
 204 encode heat shock proteins (e.g., *Hsp23*, *Hsp26*, *Hsp27*, *Hsp70Bb*), which was significantly muted in
 205 flies with ER4d neurons silenced. Other differentially-expressed genes following ER4d inhibition in-
 206 clude the enzyme tyrosine hydroxylase (*ple*), which catalyzes the rate-determining step in dopamine

207 synthesis [36]; the serine protease homolog *scarface* [37]; and several genes of unknown function.

208 We next asked how transcriptional changes following simultaneous activation of ER2 and ER4
 209 compared to manipulations of the individual neuron populations. When ER2 and ER4 neurons were
 210 activated simultaneously, we once again observed high reproducibility with replicate samples within
 211 treatments being clustered together and flies with activated ER2/ER4 neurons forming a cluster dis-
 212 tinct from both genotype and light controls (Fig. 3A). Using a false discovery rate-corrected cutoff
 213 of $p < 0.01$, we observed that ER2/ER4 activation resulted in roughly twice as many transcriptional
 214 changes as did ER2 activation alone (198 upregulated and 82 downregulated).

215 A GO analysis of the upregulated genes was dominated by metabolic pathways related to amino
 216 acid metabolism as well as pentose-phosphate- and NADPH-generating processes, with additional en-
 217 richment of biosynthetic and glycogen-metabolism-related pathways (Fig. 3B). We also observed an
 218 upregulation of several known lifespan-modulating genes involved in one carbon metabolism, includ-
 219 ing *Cbs*, *Gnmt*, *Glyp*, *Glys*, *Nmdmc*, and *Zw*. Genes involved in starvation responses and lipid and
 220 carbohydrate catabolism were significantly downregulated (Fig. 3C), which was in contrast to the
 221 upregulation of these pathways that we observed when ER2 neurons were activated alone. Notably,
 222 rather than recapitulating the effects of either of the previous two manipulations, or impacting a set
 223 of genes that was effectively a union of the two, the transcriptional profile of ER2/ER4-activated flies
 224 was distinct from those of both ER2 activation and ER4d inhibition, even where it impinged on similar
 225 pathways (Fig. 3D-E).

226 **EB ring neurons modulate nutrient-state signaling without corresponding** 227 **behavioral and physiological outputs**

228 Manipulations of ER2 and ER4d neurons alone and in combination produced markedly divergent
 229 transcriptional profiles in head tissues (Fig. 3C-D). This is at odds with a strict application of the
 230 I1-FFL circuit model [30], which would predict that these three interventions ultimately converge
 231 on a common downstream state. Our data are instead consistent with functionally distinct activity
 232 patterns across ER2 and ER4 populations encoding different interpretations of sensory stimuli or
 233 internal physiological signals, in keeping with the ellipsoid body's established role as an integrative
 234 center. If each manipulation imposes a distinguishable internal state, we reasoned that those differences

might be reflected not only in gene expression but also in the behavioral or physiological programs those states normally coordinate. We therefore asked how ER2 activation, ER4d inhibition, and ER2/ER4 co-activation affected feeding, energy storage and mobilization, and organismal performance, phenotypes expected to track changes in nutrient state, motivational drive, and overall physiological condition.

Feeding behavior does not explain nutrient-state transcriptional signatures

Reduced nutrient intake, and homeostatic hunger drive *per se*, slow aging in *Drosophila* [38, 13], and GO analyses of differentially-expressed genes in flies with activated ER2 and ER2/ER4 neurons identified pathways indicative of altered nutrient state and metabolism. We therefore asked whether the effects of neuronal activation might include changes in feeding that indicate a hunger state or that underlie the observed transcriptional differences. To assay for feeding changes, we measured the volume of food intake using a consumption-excretion (“con-ex”) assay [39]. Flies were placed on food containing a blue dye tracer for 24 hours under the same light regime used for lifespan assays, after which we quantified the amount of dye deposited on vial walls. Because flies can behaviorally and physiologically adapt to chronic neuron manipulation [13], we tested light-naïve flies as well as flies that had been pre-treated with light for one week or one month, to evaluate changes over a significant portion of the lifespan.

We found no clear evidence that activation of ER2 neurons alone or ER2/ER4 neurons in combination produces consistent changes in food consumption. Except for males of advanced age, ER2 neuron activation resulted in no significant effects on feeding. Naïve male and female ER2>Chr flies excreted amounts of blue dye statistically indistinguishable from genetic controls (Fig. 4A, left panels for each sex). After one week of light pre-treatment, individuals from the same cohort of flies were randomly distributed among new vials and tested again (Fig. 4A, center panels). We again observed no light-driven changes in feeding by ER2>Chr males or females. Following a month of light pre-treatment, males, but not females, exhibited a significant reduction of feeding, as evidenced by a statistically significant interaction effect (Fig. 4A, right). There were also no consistent effects of ER2/ER4 neuron activation on feeding, as the interaction term was not statistically significant for males or females that were light-naïve (Fig. 4B, left panels for each sex) or given one week (Fig. 4B, center) or one month (Fig. 4B,

right) of pre-treatment. We observed a nominally significant increase in ER2/ER4-activated female feeding after one week ($p = 0.041$), but this effect did not survive correction for multiple comparisons. These results suggest that ER2 and ER2/ER4 activation do not robustly regulate feeding.

Lipid stores and starvation resistance do not correlate with transcriptional signatures of ER2 and ER2/ER4 activation

If the starvation-like transcriptional signature induced by ER2 activation reflects a true nutrient-deprived physiological state, we might expect flies to exhibit hallmarks of nutrient deprivation, including reduced triglyceride (TAG) stores as well as altered starvation resistance and activity levels. After one week of ER2 activation to match RNA-seq treatment, TAG levels were unchanged in females but significantly reduced in males (Fig. 4C), whereas ER2/ER4 activation significantly reduced TAG in both sexes (Fig. 4D), contrary to the observed decrease in starvation-response genes. We then examined if TAG differences predicted survival under nutrient deprivation. Flies of both sexes were pre-treated with light for one week to match TAG assay conditions before being flipped onto 2% agar, a hydrating but non-nutritive medium. Females of both genotypes were more sensitive to starvation than controls (Fig. 4E-F, left panels). In males, starvation outcomes diverged despite overlapping effects on TAG: ER2>Chr males were only modestly sensitized to starvation compared to their dark controls, an effect driven by a long tail in control survival. By contrast, ER2/ER4>Chr males were significantly sensitized (Fig. 4E-F, right panels). That starvation sensitivity was not associated with reduced TAG was especially notable in ER2>Chr females, in which longevity extension and starvation-associated transcriptional signatures occurred without detectable TAG depletion; nevertheless, they were accompanied by reduced starvation resistance. Together, these findings indicate that EB neuron activation can alter lipid storage and starvation resistance in ways not explained by food intake and not simply predicted by inferred nutrient state, suggesting that neural activation influences nutrient-state signaling independent of canonical metabolic and physiological outputs.

Negative geotaxis reveals delayed, sex-specific effects of ring neuron manipulation

Ring neurons have established roles in locomotor control [16, 19], so we next asked whether an evoked motor assay might reveal differences among neural manipulations that measures of baseline feeding, TAG storage, and starvation resistance did not. To assess this, we tested negative geotaxis, a stereotyped climbing response widely used to measure age-sensitive locomotor performance and neuromuscular function in *Drosophila* [40, 41]. Flies were assayed using our custom “D-Drop” platform, which drops a chamber containing multiple single-fly lanes and uses video tracking to quantify several characteristics of flies’ subsequent climbing behavior. To gauge climbing speed and overall robustness, we focused on maximum height reached 9 sec after drop and total upward distance climbed over 30 sec. Light-naïve flies were omitted, as flies were not optogenetically stimulated during this assay. ER2>Chr flies of neither sex exhibited a significant genotype $\times$ light interaction for either metric after one week or one month of activation (Fig. 5A). ER2/ER4 activation likewise produced no detectable effect after one week of pre-treatment in either sex (Fig. 5B, left panel for each sex). After one month of pre-treatment, however, female ER2/ER4>Chr flies outperformed controls in both height at 9 sec and total upward distance (Fig. 5B). Males showed a modest decrease in climbing ability (Total Up $p = 0.045$), but this effect did not survive multiple testing correction.

We observed that green light *per se* shortens lifespan (Fig. 1C) and causes molecular stress responses (Fig. 2D) in both control and GtACR1 flies, and that ER4d neuron inhibition may buffer these stress responses. We therefore asked if the effects of green light stress manifest physically as reduced negative geotaxis and if ER4d inhibition would promote endurance against this purported stress. In females, we saw a modest trend toward reduced climbing ability after one week of ER4d inhibition (Total Up $p = 0.035$); however, after multiple testing correction, there were no detectable effects at either time point (Fig. 5C). In males, by contrast, ER4d inhibition improved both climbing metrics after one week pre-treatment, and this advantage persisted after five weeks of pre-treatment. ER4d activation, by comparison, produced only nominally significant improvements in climbing in either sex prior to correction (Total Up female $p = 0.040$, male $p = 0.046$) (Fig. S2A).

Overall, negative geotaxis did not clearly resolve the functional differences between ER2 and ER2/ER4 manipulations. No effects were evident at the one-week time point corresponding to our

transcriptional analysis, indicating that the early molecular states induced by these perturbations are not accompanied by parallel changes in evoked climbing behavior. Divergent effects emerged only later in ER2/ER4 females, but whether these represent delayed consequences of the earlier transcriptional programs remains unclear. By contrast, ER4d inhibition produced a robust and sustained improvement in male climbing performance. Because comparable transcriptomic data are not available for males, we cannot determine whether this sex-specific phenotype reflects a shared molecular program with different behavioral output or a distinct underlying mechanism. Though TAG levels and starvation resistance were decreased in some conditions, our ring neuron manipulations did not otherwise significantly alter baseline behavioral patterns or produce gross nutritional or performance deficits. Transcriptional patterns characterized by GO do not map onto predictable, detectable differences in these common health and behavior readouts, suggesting that lifespan extension is not a secondary consequence of changes in these phenotypes.

ER2, ER4d, and ER2/ER4 activation produce distinct valence states

Our results suggested that EB ring neurons act instructively, imposing distinct internal states whose immediate consequences are separable but whose downstream effects converge on lifespan regulation. In natural environments, ring neuron activity is engaged by behaviorally relevant stimuli, including cues that drive discrimination of nutritive value [42] or avoidance of threatening conditions [16]. Because standard husbandry conditions are relatively benign and homogeneous, however, the motivational consequences of experimentally imposed EB states may be muted under baseline conditions. We therefore next asked whether these manipulations differ in valence, reasoning that their attractive or aversive properties might clarify the functional logic by which EB circuits influence lifespan.

We first measured valence using the FLIC assay, which permits flies to associate optogenetic stimulation with self-initiated interactions with liquid food at a defined well [43]. In this paradigm, individual flies were presented with a choice between a water well and a 10% sucrose well, and closed-loop optogenetic stimulation (red light for CsChrimson or green light for GtACR1) was triggered selectively by contact with the sucrose food. Each “trigger” fly was paired with a yoked control that received identical light stimulation determined solely by the behavior of the trigger fly, without relationship to its own feeding behavior (Fig. 6A). This design allowed us to ask whether activation or inhibition

of specific EB neuron populations altered the perceived reward value of the sucrose-associated food while accounting for non-associative consequences of light exposure (increased hunger, for example). As with our behavioral assays, we performed these experiments both in light-naïve flies and in flies pre-exposed to light for one week or one month. Because FLIC data can be used to extract several features of feeding behavior, it allowed us to assess the valence of each manipulation using multiple behavioral dimensions. Each FLIC signal over threshold was scored as an “interaction” with the liquid food, and these data were further parsed into the number of discrete well visits (“events”) and the duration of those visits (“median duration”) [44]. Increased sucrose interactions, more frequent visits, or longer visit durations were interpreted as evidence of positive valence, whereas rapid abandonment of the trigger well or subsequent avoidance was taken to indicate negative valence. Due to the sophistication and limited throughput of this assay, and because our transcriptomic analyses were performed in females, we restricted this component of our study to females to enable direct comparison between observed valence and underlying transcriptional states.

Again, we first tested ER2 activation, prompted by the observation that ER2>Chr animals exhibited transcriptomic signatures consistent with nutrient stress and starvation-like state changes. If so, we reasoned that activation of ER2 neurons might be negatively valenced or might alter the motivational salience of a nutritive stimulus, respectively. Indeed, light-naïve ER2>Chr trigger flies showed a marked reduction in interactions with the sucrose well relative to yoked controls, indicating that contact with the trigger well was strongly aversive despite the inherent attractiveness of sucrose (Fig. 6B, left). This effect was accompanied by a significant reduction in event duration (Fig. S2A, left), suggesting that flies rapidly disengaged from the well once stimulation was triggered. Notably, however, ER2>Chr flies did not reduce the number of visits to the sucrose well, implying that ER2 activation did not simply condition avoidance of the well itself (Fig. S2A, right). Instead, these data suggest a more acute aversive state that promotes premature disengagement while leaving repeated approach behavior largely intact. This trend remained evident after one week (Fig. 6B; $p = 0.035$ uncorrected) and one month (Fig. 6B, right) of light pre-treatment, indicating that the negative valence associated with ER2 activation does not substantially diminish with chronic exposure.

Because the transcriptional profile induced by ER2 activation was associated with genes involved in starvation response, we considered whether the aversive effect of ER2 stimulation might obscure

373 a parallel increase in motivation to feed. In principle, if ER2 activation enhanced food drive while
 374 simultaneously imposing negative valence, then the immediate aversive response of trigger flies could
 375 preclude increased sucrose consumption. To test this possibility, we modified the yoked design to
 376 include a single, wild-type trigger fly and two yoked animals, one ER2>Chr and one genetic control
 377 (Fig. S2B, left). Under this configuration, both yoked animals experienced light delivery uncoupled
 378 from the food and at a frequency determined by normal feeding of the wild-type trigger fly. If ER2
 379 neuron activation increased hunger, we would expect the yoked ER2>Chr flies to approach or interact
 380 with the sucrose well more often than yoked control animals. Despite equivalent stimulation, however,
 381 yoked ER2>Chr flies did not show increased sucrose consumption relative to yoked controls (Fig. S2B,
 382 right), arguing against the idea that ER2 activation induces a hunger phenotype, which is consistent
 383 with our feeding data described above.

384 We next sought to evaluate the behavioral consequences of ER4d inhibition. Interpretation of
 385 these experiments, however, was again complicated by the green light required for GtACR1 activation.
 386 In our assay conditions, abrupt green-light delivery appeared to be intrinsically aversive or startling,
 387 producing reduced interactions and shorter event durations in both ER4d>GtACR1 flies and genetic
 388 controls. Although trigger flies expressing GtACR1 in ER4d neurons displayed aversion to the sucrose
 389 well, a similar response was observed in trigger flies of the control genotype, precluding a clear assign-
 390 ment of valence to ER4d inhibition itself (Fig. S2C). Thus, under these conditions, the FLIC assay
 391 could not distinguish the effects of ER4d inhibition from the effects of light exposure.

392 Given these limitations, we instead used activation as an alternative means of probing the behavioral
 393 state associated with this population. In light-naïve flies, ER4d activation produced a behavioral
 394 signature resembling that of ER2 activation: trigger flies had fewer interactions with the sucrose well
 395 (Fig. 6C, left; uncorrected $p = 0.05$), made shorter visits (Fig. S2D, left), and showed no reduction in
 396 the number of visits (Fig. S2D, right). Thus, as with ER2 stimulation, ER4d activation appeared to
 397 promote rapid disengagement from the trigger well rather than suppressing repeated approach to the
 398 sucrose source itself. This aversive effect persisted after one week of pre-treatment (Fig. 6C, center)
 399 but was no longer evident after one month of chronic light exposure (Fig. 6C, right), suggesting that,
 400 unlike the response to ER2 activation, the negative valence associated with ER4d activation attenuates
 401 over time.

402 We next asked how flies respond when ER2 and ER4 neurons are activated simultaneously. Based
 403 on the aversive effects of activating either ER2 or ER4d alone, one might expect combined activation
 404 to produce a similar or even stronger negative response. Instead, ER2/ER4 activation in light-naïve
 405 flies increased the number of interactions with the sucrose well (Fig. 6D, left; uncorrected $p = 0.020$).
 406 Although median visit duration was unchanged (Fig. S2E, left), the frequency of visits increased
 407 (Fig. S2E, right), indicating that stimulation reinforced engagement with the trigger-associated food
 408 source. ER2/ER4-activated trigger flies continued to interact more with the sucrose well than their
 409 yoked controls after one week and one month of pre-treatment; however, at these later time points,
 410 there was no longer a significant (at $p < 0.05$) genotype $\times$ light effect (Fig. 6D, center and right,
 411 respectively), perhaps due to an increase in among-fly variance. Nevertheless, ER2/ER4 flies showed
 412 a trend toward greater attraction to the well than controls at these later time points ($p = 0.06$ – 0.1).

413 Together, these data indicate that simultaneous activation of ER2 and ER4 neurons produces a state
 414 consistent with positive valence, in contrast to the aversive effects observed when either population is
 415 activated alone.

416 **EB valence is shaped by context, sex, and circuit composition**

417 To test whether the valence effects observed in the FLIC generalized beyond a feeding context, we
 418 assayed place preference in a rectangular multi-lane arena in which one half of each lane was illuminated
 419 and the other remained dark. Unlike the closed-loop FLIC, this open-loop design delivered light on
 420 a pre-programmed schedule regardless of fly position, dissociating valence from sucrose contact or
 421 response-contingent stimulation. Flies were observed during a 10-min dark acclimation period followed
 422 by a 60-min light treatment period of 5 sec on/5 sec off. Light-naïve flies of both sexes were tested,
 423 and each manipulation was evaluated at two stimulation frequencies (48 Hz to mimic FLIC conditions,
 424 and 200 Hz) to further gauge generalizability.

425 The positive valence of ER2/ER4 co-activation generalized cleanly. Both sexes significantly pre-
 426 ferred the illuminated side during treatment, an effect observed at both 48 Hz (Fig. 7B) and 200 Hz
 427 (Fig. S3A), indicating that the appetitive state reflects a stable property of the ER2/ER4-activated
 428 circuit rather than an artifact of a particular assay or stimulation regime.

429 Consistent with the FLIC results, activation of ER2 neurons in females produced a clear avoidance

of the illuminated side during the treatment period (Fig. 7C, left), indicating that the negative valence associated with ER2 activation extends beyond feeding-related behavior. This effect was also observed under a higher-frequency stimulation regime, supporting the robustness of the phenotype across light conditions (Fig. S3B, left). In males, ER2 activation produced no preference at 48 Hz (Fig. 7C, right) or 200 Hz (Fig. S3B, right). Thus, the arena revealed a sex-dependent valence for ER2 activation.

By contrast, the FLIC findings for ER4d manipulation did not generalize robustly. ER4d>GtACR1 and control flies showed indistinguishable side biases (females preferring illumination, males avoiding it; Fig. S3C), so although the open-loop design removed the light-startle confound, it did not permit an unambiguous valence assignment. ER4d activation, which had been aversive for naïve flies in the FLIC, produced variable behavior in the arena. ER4d activation produced an aversive shift in both sexes at 48 Hz (Fig. 7D, left), although the effect was stronger in males. Females exhibited reduced light preference relative to controls, whereas males preferred the unlit side. Neither sex exhibited a significant place preference at 200 Hz (Fig. S3D, left). The absence of a stable signature across conditions suggests that ER4d neurons act in a more context-sensitive or modulatory fashion than ER2 or ER2/ER4.

In summary, the behavioral consequences of EB manipulation are nonlinear and depend on circuit composition, sex, and assay context. ER2/ER4 co-activation produced a robust positive valence that generalized across sex and assay paradigms, whereas ER2 and ER4d manipulations did not yield consistently generalizable valence signatures. Together, these findings argue that EB ring neurons do not make fixed, additive contributions to motivational state, but instead generate qualitatively distinct and differentially stable internal states through combinatorial recruitment, raising the question of how such state differences relate to the common downstream outcome of extended lifespan.

ER2 neuron activation remodels metabolism in peripheral tissues

Finally, to examine how EB neuron manipulation in the brain influences gene expression and physiology in the periphery, we performed bulk RNA-sequencing and metabolomic profiling on headless bodies of female flies. We focused on ER2-activated flies to determine whether the strong starvation-response program observed in heads would manifest in the periphery. Body donors were reared and collected the same way as head donors. Bodies were separated from heads using a sieve, and five bodies

were collected for each of five samples. Again, using DESeq2 to identify genes with a significant genotype (CsChrimson, control) $\times$ light (yes, no) interaction, we identified 229 upregulated and 181 downregulated genes in peripheral tissues (experiment-wise p-value < 0.01) (Fig. S4A). Strikingly, the peripheral transcriptional response was nearly completely distinct from that observed in the head, with only 7 of 526 differentially-expressed genes shared between compartments (Fig. 8A), indicating that ER2 activation drove tissue-specific transcriptional programs.

Despite the disparity in response genes, downregulated processes were consistent with the starvation-like transcriptional profile in heads, as ER2 activation drove peripheral downregulation of genes involved in cell cycle progression and DNA synthesis and repair (Fig. 8B). These included replication factors *PolA2*, *Cdc45*, and the Cdt1 ortholog *dup* [45]; mitotic regulators *Cdk1*, *twe* (Cdc25), and *fzy* (Cdc20); kinetochore and centrosome components *Mis12*, *Sas-4* (CENPJ), and *spd-2* (Cep192); and DNA repair factors *RecQ5*, *Mlh1*, and *Pms2*. Chromatin remodeling and transcriptional factors were likewise downregulated, indicating a shift away from proliferative states.

By contrast, we did not observe an upregulation of canonical stress- or starvation-response genes in bodies. Instead, the peripheral response was characterized by electrochemical homeostasis and organelar acidification (Fig. 8C). Among the most significantly upregulated genes were more than a dozen vacuolar H⁺-ATPase subunits and accessory factors, including *Vha16-1*, *Vha68-2*, *Vha44*, *VhaAC45*, *VhaSFD*, *Vha26*, *VhaAC39-1*, *Vha55*, *Vha100-2*, and *ATP6AP2*. This V-ATPase program was not accompanied by the upregulation of autophagy or trafficking pathways. We also noted the upregulation of three inwardly rectifying potassium channels (*Irk1*, *Irk2*, and *Irk3*), as well as an enrichment of lipid-associated processes, including genes involved in fatty acid synthesis (*ACC*) and ether phospholipid synthesis (*Agps*). Notably, we also observed inductions of *Sam-S* and *Gnmt*, indicating changes in methionine flux.

Targeted metabolomic profiling of peripheral tissues using a similar experimental design, but a less restrictive experiment-wise $p < 0.05$ cutoff, revealed a broad and coordinated metabolic remodeling that reinforced our interpretation of the transcriptional response. PCA revealed that the single most influential source of variance in the data was neuronal activation, as PC1 (accounting for 41.2% of the variance) separated ER2>Chr flies exposed to light from all other samples (Fig. 9A). A total of 87 metabolite abundances were found to be significantly affected by ER2 neuron activation (Fig. 9B-C).

487 Among the 47 metabolites reduced in abundance, the dominant pattern was broad depletion of free
 488 nucleotides and nucleosides, including purines, pyrimidines, and modified nucleoside species, along-
 489 side a broad reduction in free amino acids and derivatives. Reductions in S-adenosylhomocysteine
 490 (SAH), choline, glycerophosphocholine, and acylcarnitines were also observed. Three biochemical
 491 modules emerged from the 40 significantly upregulated metabolites: a central carbon module com-
 492 prising glycolytic and TCA-cycle intermediates including pyruvate, oxaloacetate, and cis-aconitate
 493 alongside pentose phosphate pathway species; a small nitrogen species module containing glutamine,
 494 arginine, serine, and betaine; and a tryptophan-kynurenine module marked by coordinated elevation
 495 of kynurenic acid, quinolinic acid, and anthranilic acid. Pathway-level analysis using FELLA, which
 496 maps metabolite changes onto the KEGG network to identify enriched biological modules, corrob-
 497 orated these findings and indicated additional enrichment of taurine and hypotaurine metabolism,
 498 beta-alanine metabolism, and folate transport, with starch and sucrose metabolism and nucleotide
 499 sugar biosynthesis also represented (Fig. 9D).

500 Discussion

501 Here, we show that direct activation of ER2 neurons, co-activation of ER2/ER4 neurons, and inhibition
 502 of ER4d neurons are each sufficient to promote longevity in adult flies of both sexes. These effects are
 503 elicited by manipulating very small neuronal populations—in some cases, fewer than ten neurons per
 504 hemisphere [46, 15]—underscoring the sensitivity of organismal aging to defined circuit states. Our
 505 findings establish that the EB can act instructively to modulate aging rather than serving only as
 506 a permissive gate for lifespan-altering sensory information. Previous work showed that ER2 neurons
 507 are required for death perception to accelerate aging [20] and, notably, that constitutive activation
 508 of ER2/ER4 neurons using heat-gated TRPA1 modestly reduced lifespan [20]. Read together with
 509 the data reported here, these contrasting outcomes suggest that the timing and persistence of EB
 510 activity help define ER-neuron-dependent internal states and their effects on aging, consistent with
 511 the context-sensitivity of valence that we observed.

512 We also find that manipulations of different EB populations likely modulate aging through distinct
 513 molecular mechanisms. Bulk head transcriptomics after one week of treatment revealed that ER2 acti-
 514 vation, ER4d inhibition, and ER2/ER4 co-activation produced largely non-overlapping transcriptional

515 programs. The differences were not merely quantitative: where the manipulations converged on shared
 516 pathways, they often did so in opposite directions, with starvation- and catabolism-related genes up-
 517 regulated by ER2 activation but downregulated by ER2/ER4 co-activation. Rather than resolving
 518 immediately to a common state, then, distinct activity patterns across ER2 and ER4 populations
 519 appear to be proximally interpreted as distinct internal states that each extend lifespan.

520 Our experiments also revealed, perhaps fortuitously, that ER4d inhibition may enhance the abil-
 521 ity to buffer moderate stress. Green light alone shortened lifespan and broadly induced heat-shock
 522 proteins (e.g., *Hsp23*, *Hsp26*, *Hsp27*, *Hsp70Bb*) in both control flies and flies expressing GtACR1.
 523 ER4d silencing produced few detectable genotype-specific changes apart from attenuating this stress
 524 signature. Because temperatures in our ventilated rigs were not meaningfully elevated above those of
 525 dark-aged controls, heat is unlikely to explain these effects. Fly vision is more sensitive to green than to
 526 red light [47], and certain wavelengths can exert physical effects on the brain [48] and shorten lifespan
 527 [49]; the intensity and duration of exposure required for deep brain penetration in our experiments
 528 [50] likely rendered green light a meaningful stressor. Whether the attenuated molecular response in
 529 ER4d-silenced flies actively contributed to their longer lifespan, or instead reflects a reduced stress
 530 burden, remains to be determined, as does whether a comparable mechanism operates under more
 531 benign conditions.

532 The transcriptional data also point to specific pathways that may link EB activity to aging-
 533 regulatory physiology. Following ER2 activation, the induction of *foxo*, *Thor*, *InR*, *dilp6*, and *Cbs*
 534 suggests a coordinated engagement of nutrient-sensing, proteostatic, and sulfur-amino-acid metabolic
 535 processes previously implicated in lifespan control. This state does not, however, resemble a canonical
 536 starvation program: nutrient-responsive IIS components were induced alongside ribosome biogenesis
 537 and translational programs that are not ordinarily co-regulated in this way [51], suggesting that di-
 538 rect EB activation may uncouple pathways normally coordinated by nutrient availability. Because
 539 prior work showed that ER2/ER4 neurons regulate *dilp3* and *dilp5* expression in insulin-producing
 540 cells during death perception [20], one plausible hypothesis is that EB-dependent internal states are
 541 translated into systemic aging effects through downstream neuroendocrine centers, though the present
 542 data do not establish this mechanism directly, and defining the relevant effector pathways will require
 543 future work.

544 The internal states established by EB manipulation were molecularly specific yet were not accom-
 545 panied by the behavioral or physiological changes those states would ordinarily predict. GO analysis
 546 revealed opposite transcriptional indicators of nutrient availability in the head: starvation-response
 547 processes were upregulated by ER2 activation (Fig. 2B) but downregulated by ER2/ER4 activation,
 548 which instead enriched for biosynthetic and storage-associated processes including amino acid syn-
 549 thesis and glycogen metabolism (Fig. 3C), suggestive of a nutrient-replete state. Neither signature
 550 was matched by a change in feeding. The ER2 signature could have reflected reduced intake or a
 551 hunger-like state that should itself promote feeding, yet we observed neither; the nutrient-replete-like
 552 ER2/ER4 signature was likewise unaccompanied by any feeding change. These transcriptional states
 553 are therefore unlikely to arise secondarily from altered consumption, and they argue against dietary
 554 restriction as the basis of lifespan extension. The same dissociation extended to other coarse phe-
 555 notypes: despite unchanged feeding, TAG stores were reduced in ER2 males and in both sexes after
 556 ER2/ER4 activation, yet these changes tracked neither lifespan extension nor starvation resistance,
 557 and negative geotaxis revealed no early correlate of the one-week transcriptional states. Together,
 558 these results indicate that EB-dependent regulation of aging is not readily explained by gross changes
 559 in feeding, lipid storage, or locomotor performance.

560 The absence of feeding and physiological correlates did not, however, mean that EB manipulation
 561 was motivationally silent; each manipulation instead imposed a distinct, dissociable motivational state.
 562 In the FLIC assay, ER2 activation in females was aversive, promoting rapid disengagement from the
 563 sucrose well without suppressing re-approach, whereas ER2/ER4 co-activation was positively valenced,
 564 increasing both the number of interactions and the frequency of return without prolonging individual
 565 feeding bouts. In a feeding-independent place-preference assay, the positive valence of ER2/ER4 co-
 566 activation generalized across sex and stimulation regimes. By contrast, aversive states were more
 567 context-dependent: the negative valence of ER2 activation was restricted to females but held across
 568 stimulation regimes, while that of ER4d activation generalized across sex but only emerged under
 569 different stimulation regimes. Together, these findings indicate that EB ring neurons do not contribute
 570 fixed, additive aversive or appetitive signals; rather, the motivational meaning of EB activity depends
 571 on population combination, assay context, sex, and stimulation conditions.

572 A particularly informative consequence of this combinatorial structure is that manipulations of

opposite valence both extend lifespan: ER2 activation, which was aversive in females, and ER2/ER4 co-activation, which was appetitive, each reliably increased survival in males and females. This is difficult to reconcile with models in which longevity is mediated by a particular motivational valence, or by any single downstream behavioral or physiological state that valence would normally engage. It is also difficult to reconcile with a strict reading of the incoherent feedforward circuit in which ER2, ER4m, and ER4d neurons are thought to participate [30], under which ER2 activation, ER2/ER4 co-activation, and ER4d inhibition should converge on a common downstream state. This model is at odds with the divergence we observed in transcriptional profile, valence, and downstream physiology, although we cannot exclude convergence on a shared terminal mechanism farther downstream of the EB. A more parsimonious interpretation is that EB circuits influence aging through population-level configurations whose downstream consequences depend on the particular combination of neurons engaged, consistent with the EB's established role as an integrative rather than relay structure. In this view, lifespan extension is not the signature of a single privileged EB activity pattern but a property of multiple distinguishable circuit states, each recruiting partially distinct downstream programs that nonetheless converge on slower aging, with the transcriptional divergence we observed implying distinct molecular routes to a shared outcome.

Molecular profiling of peripheral tissues revealed that ER2 activation drove coordinated systemic remodeling beyond the brain. In bodies, ER2 activation suppressed proliferative and DNA repair programs while upregulating a vacuolar H^+ -ATPase-dominated electrochemical homeostasis program not observed in heads, indicating tissue-specific rather than uniform physiological responses. That this V-ATPase program was not accompanied by autophagy or trafficking pathways suggests a decoupling of organellar acidification from degradative and secretory functions. The accompanying upregulation of inwardly rectifying potassium channels may help maintain cell membrane potential during increased organellar proton flux, while the enrichment of fatty acid and ether phospholipid synthesis genes suggests a remodeling of cellular and organellar membranes to support increased ion transport.

Metabolomic profiling reinforced this interpretation. The depletion of free nucleotides and amino acids is consistent with the reduced proliferative burden evident in our transcriptional data, while reductions in choline, glycerophosphocholine, and acylcarnitines may reflect increased membrane remodeling activity. Among upregulated metabolites, the accumulation of glycolytic and TCA-cycle

intermediates, together with glutamine and arginine, is consistent with reduced consumption of these species by growth-related processes. Serine and betaine serve as methyl donors, and their elevation, alongside depletions in SAH and choline and the upregulation of both *Sam-S* and *Gnmt* in head and body, indicates a restructuring of methionine cycle flux. Although none of the core kynurenine pathway genes were upregulated in our RNA-seq dataset, the coordinated elevation of kynurenic acid, quinolinic acid, and anthranilic acid could account for the depleted tryptophan signature. The convergence of transcriptomic and metabolomic data indicates that activation of the small population of ER2 neurons imposes an orchestrated, tissue-specific remodeling of peripheral physiology that is distinct from canonical starvation or dietary restriction.

Our finding that discrete or combined populations of ellipsoid body ring neurons can instructively modulate lifespan by generating distinct internal states carries potentially significant implications for understanding the neural basis of aging in humans. The *Drosophila* central complex, of which the ellipsoid body is a core component, shares strong neuroanatomical and functional homology with the vertebrate basal ganglia [14]. Both structures serve as integrative hubs that process sensory information, coordinate motor output, and critically contribute to the generation of motivational and affective states. In particular, limbic basal ganglia output pathways, including ventral pallidal circuits, encode opposing appetitive and aversive motivational value and promote approach or avoidance behaviors according to internal state and motivational context [52]. Our findings raise the possibility that a similarly general principle applies to aging: circuit configurations that impose distinct motivational or physiological states may converge on long-term effects on organismal maintenance and survival. More generally, these results suggest that lifespan may be modulated not only by what animals perceive, but by how neural state transforms the meaning of those perceptions into coordinated physiological output. If analogous circuits in the human brain similarly encode valence states that influence systemic physiology, this would provide a mechanistic framework for understanding associations between psychological states, chronic stress, and aging in humans.

Materials and Methods

Fly stocks and husbandry

Eggs were collected from grape juice agar as described in Gendron et al., 2023, and 32 μ L of eggs were aliquoted into bottles containing 25-50 mL of CT larval growth medium [53]. Channelrhodopsin-expressing flies and their genetic controls were reared and maintained in dim 12:12 light:dark conditions at 25° C and 60% humidity. Unless otherwise specified, flies were collected 0-1 days after eclosion, transferred to 10% sucrose/10% yeast (SY10) medium, and allowed to mate for 2-3 days. Mated flies were then sorted by sex onto all-trans retinal (500 μ M ATR) SY10 medium, 20/vial, under brief CO₂ anesthesia. Adult flies were flipped onto fresh food every 2-3 days until completion of experiments. Dark controls were flipped in a dark room to minimize incidental channel activation. The following stocks were used, and obtained from the Bloomington Drosophila Stock Center: white Canton-S (WCS), UAS-CsChrimson (BDSC #55315), UAS-GtACR1 (BDSC #92983), UAS-20X-GFP (BDSC #32194), ER2-GAL4 (BDSC #48343), ER4d-GAL4 (BDSC #48487), ER2/ER4-GAL4 (BDSC #75997), and BDSC #39805 (broader EB).

Optogenetic assays

After being sorted onto ATR food, flies were placed in a dark box for at least 1 day to allow enough time for all-trans retinal uptake before light stimulation. Flies used for lifespan assays or 1-week or 1-month assay time points were then transferred to optogenetic rigs. Dark controls remained in a dark box in the same incubator. Light-naïve flies for behavioral assays were maintained for 1 week in the dark box to age-match 1-week light-treated flies. Our optogenetic rigs house 48 vials between 2 LED light bars, and rig walls are mirrored to facilitate consistent light distribution across vials. Rigs and behavioral assay equipment use 627 nm (red) or 530 nm (green) Luxeon LEDs. Control boards allow the experimenter to set light intensity, duration, and other stimulus parameters. Vial position was randomized within each optogenetic rig and dark box.

651 Lifespan assays

652 Each treatment comprised 5-10 vials of 20 flies each (see Fig. 1 caption for populations per experiment).
 653 Flies were 3-9 days old when entering light. Experimental and control genotypes were established in
 654 parallel and were generally age-matched. Flies were exposed to 12 V (red) or 17-20 V (green) light
 655 during ZT 0-12. Unless otherwise specified, stimuli were delivered continuously throughout this interval
 656 at 40 Hz with a 32% duty cycle. Male ER2>Chr flies received an intermittent stimulation schedule
 657 consisting of 6 min on and 12 min off cycles during ZT 0-12. Rigs and dark boxes were maintained in
 658 incubators set to 25° C and 60% humidity.

659 Starvation assays

660 Flies were 3-4 days old when placed into light for 1 week of pre-treatment before starvation assays.
 661 After pre-treatment, flies were removed from ATR medium, sorted into 10 vials of 10 flies, and placed
 662 onto 2% agar, which provides water but not nutrition. Anti-fungal agent Tegosept was omitted from
 663 starvation medium to avoid introducing a nutritive solvent (EtOH). Flies were treated with red light
 664 for the duration of the assay and flipped onto clean agar every 2-3 days. Surviving flies were counted
 665 every few hours. Counting frequency increased as the experiment progressed, as starving vials often
 666 go extinct within a short window. Due to the high frequency of census taking, starvation flies were
 667 not kept in an incubator, but at room temperature (21.1-23.3° C) and ambient humidity.

668 Survivorship analysis

669 For lifespan and starvation survival analysis, we performed pairwise comparisons of dark and light-
 670 treated survivorship curves using the DLife software [28] and R. P-values were obtained by log-rank
 671 analysis of color-coordinated pairs in Fig. 1 and Fig. 4.

672 Brain dissections

673 Brain dissections, immunostaining, and imaging were performed largely as described previously [20],
 674 with minor modifications. Driver (GAL4) lines were crossed with UAS-20X-GFP. Adult female brains
 675 were dissected in phosphate buffer saline (PBS) and fixed for an hour at room temperature in PBS

676 containing 4% paraformaldehyde. Brains from ER2, ER2/ER4, and BDSC #39805 drivers were trans-
 677 ferred to a glass slide with Vectashield mounting medium (ZH0219), sealed with a coverslip and nail
 678 polish, and immediately imaged. Because ER4d comprises relatively few neurons, ER4d>20X-GFP
 679 brains were immunostained for GFP before being similarly mounted for imaging.

680 Immunostaining

681 ER4d-GFP brains were transferred to a microcentrifuge tube and quickly washed 3 times with 1 mL
 682 of PBS with 0.1% Triton-X (PBS-T). Brains were then given 3 longer 1 mL PBS-T washes, with
 683 tubes placed on a low-speed orbital shaker at room temperature for 20 min for each wash. PBS-T was
 684 removed, blocking solution was added (5% normal goat serum), and tubes were returned to the orbital
 685 shaker for 30 min at room temperature.

686 Blocking solution was removed and a 1:500 dilution of rabbit anti-GFP (Sigma-Aldrich SAB4301138)
 687 in 5% normal goat serum was added. Tubes were placed on an orbital shaker at 4° C for 2 days. On
 688 day 3, the anti-GFP solution was removed and brains were given 3 quick and 3 long PBS-T washes as
 689 described above. PBS-T was removed and a 1:1000 dilution of anti-rabbit Alexa Fluor 488 (Invitrogen
 690 A-11008) in 5% normal goat serum was added. Tubes were covered in foil and placed on an orbital
 691 shaker at 4° C overnight, after which the secondary antibody was removed and brains were given 3
 692 quick and 3 long PBS-T washes. Brains were then mounted for imaging as described above.

693 Imaging and analysis

694 An Olympus FLUOVIEW FV3000 confocal microscope was used to image brains for Fig 1. Images
 695 were acquired with a 20× (0.75 NA) objective lens at 1024 × 1024 pixels with a step size of 3.0
 696 μm. Imaging files were analyzed using Fiji. Selective slices were combined using SUM projection.
 697 Brightness and contrast were adjusted similarly across treatments for visualization purposes.

698 Consumption-excretion assay

699 Con-ex experiments were performed as previously described [39], with the addition of ATR (250 μM
 700 final) to the dyed food (FD&C Blue No. 1). Light-naïve flies were sorted 10/vial instead of 20/vial
 701 for overnight ATR pre-treatment, to allow for direct flipping into con-ex vials without anesthesia the

702 following day. From an initial cohort of 120 flies, 100 were chosen for each time point (10 vials). The
 703 remainder were treated identically to the analyzed experimental flies (including during assays) and
 704 were maintained to compensate for possible age-related attrition at later time points. Flies were 3-4
 705 days old for the first (light-naïve) timepoint, to match standard lifespan conditions. After 24 hours,
 706 dyed food caps were removed, and flies were returned to standard ATR SY10 medium (20/vial) and
 707 placed back in the light rig or the dark box. Dye deposited on walls was washed by vortexing 3 mL
 708 of DI water in each vial. Absorbance was measured as described by Shell et al., 2018. For subsequent
 709 time points, remaining flies were re-mixed on a cold pad, and 100 per condition were selected for
 710 analysis as above. 1-week time points were performed after 7 days in light (i.e. 6 days after end of
 711 naïve time point). Male ER2/ER4 1-month time point was performed after 34 days in light; all others
 712 were performed after 28 days in light. Blue excrement did not persist until subsequent time points.
 713 Vials in which one or more flies died or became stuck during the assay were excluded from analysis.

714 TAG assay

715 Three- to 5-day old flies were placed in red light rig or dark box for 1 week, then frozen at -80° C.
 716 Flies were homogenized in groups of 10 (except for ER2/ER4 females, which were in groups of 5) in
 717 150 μ L PBS with 0.05% Triton X. TAG was measured using the Infinity Triglyceride reagent (Thermo
 718 Electron Corp.) Ten biological replicates were analyzed.

719 Negative geotaxis assay

720 Negative geotaxis was assayed using equipment and tracking software developed by our lab (D-Drop
 721 Deluxe and DTrack). Flies were maintained as in lifespans (12 V red light or 19.5-20 V green light)
 722 and given 1 week (6-8 days) or 1 month (21-34 days) of light treatment starting at 3-8 days of age,
 723 after 2-5 days of mating. D-Drop does not have integrated optogenetic LEDs, making this the only
 724 behavioral assay without concurrent neuron activation/inhibition (therefore no naïve time point was
 725 included). For assay, flies were anesthetized on a cold pad and individually transferred into lanes of
 726 custom 12-lane or 14-lane chambers, then placed at 25° C in ambient light or in a dark box for 10-25
 727 minutes to recover. Chambers were then loaded onto the "D-Drop Deluxe," an instrument that drops,
 728 illuminates, and records videos of single chambers. Each chamber was dropped 3 consecutive times

and recorded for 30 sec after each drop. After recording, flies were given an additional 30 sec to recover (60 total sec between drops). Light-treated flies were dropped under ambient light conditions, with a ring light (for recording) contributing additional light. Dark controls, though kept light-naïve until the assay, were also subjected to this ring light during recording. Height reached by 9 sec after drop and total up distance climbed in 30 sec after drop were averaged across replicates. Interaction (light $\times$ genotype) p-values were derived from 2-way ANOVAs. Benjamini-Hochberg FDR correction was applied within each sex and driver and across time points and climbing metrics.

Fly liquid-food interaction assay

Flies were assayed by the Fly Liquid-food Interaction Counter (FLIC) developed by our lab [43].

Two-choice FLIC *Drosophila* Feeding Monitors (DFMs, Sable Systems International, model DFMV3) were used, with one well containing 10% sucrose (w/v) in DI water and the other containing DI water alone. Each DFM contained 6 chambers, and genotypes were mixed within DFMs, although 2- or 3-way yoked pairing was maintained across neighboring chambers. Flies were anesthetized on a cold pad, aspirated into chambers, and given a few minutes to recover before recording began. Interactions were recorded for 6 hours between ZT 3 and 11 to avoid crepuscular feeding peaks. DFMs were staged in enclosed environments maintained at 25° C.

Flies were allowed to mate for 2-6 days before sorting onto ATR, and started light treatment (or aging in dark box) at 3-7 days post-eclosion. Light-naïve flies were aged for 7-12 days in dark before assays. Flies assigned to the 1-week time point received 7-14 days of pre-treatment before testing. One-month time points were pre-treated for 28-29 days.

Data were pooled from three replicates and analyzed using R code developed in-house: https://github.com/PletcherLab/FLIC_R_Code (minimum feeding threshold = 10, event gap = 5, tasting threshold = (0, 10)). Flies with no feeding signal, or whose baseline signal was unstable, were omitted from analysis. If a "trigger" fly engaged in 0 sucrose well interactions, both it and its yoked control were omitted from analysis. Sucrose interaction p-values were derived from the genotype $\times$ linkage (where linkage = trigger fly or yoked control) interaction term of a linear mixed model (Interactions Genotype * Linkage + (1|Group), lmer) with yoked pairs modeled as a random intercept. Events and median durations were analyzed similarly. Benjamini-Hochberg FDR correction was applied within

each sex and driver and across interaction time points.

Place preference assay

Young light-naïve flies (< 21 days post-eclosion) were aspirated into individual lanes of a multi-lane rectangular arena (Fig. 7A). Flies were given 10 minutes to acclimate to the arena in the dark, after which one half of the transparent arena floor was illuminated for 60 min (5 sec on/5 sec off). GtACR1 flies and their genetic controls were exposed to green light at 24 V/40 Hz/32% duty cycle. CsChrimson flies and their genetic controls were exposed to red light (18 V) at either 48 Hz/32% duty cycle (meant to mimic FLIC light parameters) or 200 Hz/20% duty cycle. Control and experimental genotypes were alternated between lanes. The illuminated side of the arena was varied between replicates. Data were pooled across 2-3 replicates.

RNA sequencing data acquisition

For bulk head and body transcriptomics, 3- to 6-day old female flies were placed into optogenetic rigs or maintained in the dark in the same incubator for 1 week, after which they were flash-frozen in liquid nitrogen. The same light parameters were used as for lifespans above. Flies were combined within conditions and vortexed to remove heads from bodies. Heads and bodies were sorted using a sieve, counted on dry ice in a petri dish, and placed into chilled bead lysis tubes. Ten heads or 5 bodies (ER2 only) were grouped per sample, and 5 samples were sequenced per condition. RNA was extracted using the New England BioLabs Inc. (NEB) Monarch Total RNA Miniprep Kit (T2010S). Samples were lysed in 800 μ L of the kit's lysis buffer at 6.5 m/s for 15 seconds in a FastPrep-24 homogenizer (MP Biomedicals). Bulk RNA was extracted according to manufacturer's instructions (including the optional DNase treatment) and sent to Admera Health LLC for library preparation using polyA selection and the NEBNext Ultra II Directional RNA Library Prep Kit. Libraries were sequenced on an Illumina platform with 2 $\times$ 150 bp paired-end reads at a depth of 40 M reads per sample (20 M per direction).

RNA sequencing data processing and statistical analysis

Raw sequencing counts were analyzed using the DESeq2 package (v. 1.48) in R [31]. We used a 2×2 factorial design with treatment (light vs. dark) and genotype (experimental vs. control) as factors, modeling gene expression as `Treatment + Genotype + Treatment:Genotype`. Dark and control were used as reference levels. We pre-filtered for genes with counts ≥ 10 in at least 3 samples. Differential expression was assessed using DESeq2’s default Wald test with Benjamini-Hochberg FDR correction. Genes with $\text{padj} < 0.01$ were considered differentially expressed.

Heatmaps of log2-transformed normalized counts of differentially expressed genes were generated using the pheatmap R package (v 1.0.13) [54], with hierarchical clustering (Euclidean distance, complete linkage) and row-wise z-score scaling.

Gene Ontology (GO) enrichment analysis was performed using the clusterProfiler package (v. 4.16) in R [55]. Differentially expressed genes ($\text{padj} < 0.01$) were separated into upregulated and downregulated sets and tested independently for enrichment of Biological Process terms, using the pre-filtered gene set as the background universe. Enrichment significance was assessed with a Benjamini-Hochberg FDR adjusted p-value cutoff of 0.05 and a q-value cutoff of 0.2. Redundant GO terms were collapsed using clusterProfiler’s simplify function with a semantic similarity cutoff of 0.7. Rather than retaining the most statistically significant term in each cluster, we retained the most specific term, defined as the term with the greatest number of ancestors in the GO Biological Process hierarchy.

Metabolomics data acquisition, processing, and statistical analysis

Three-day-old female flies were treated and harvested as described for RNA sequencing. Six samples of 30 whole female bodies were sent to the University of Washington’s metabolomics core for analysis. Approximately 10 mg of body tissue samples were homogenized in cold Milli-Q water and metabolites were extracted with methanol containing isotopically labeled internal standards (^{13}C -glucose and ^{13}C -glutamic acid). Samples were vortexed, incubated at -20°C , sonicated, and centrifuged at 14,000 rpm for 15 minutes at 4°C . Supernatants were collected, dried under vacuum, and stored at -80°C . Prior to analysis, dried extracts were reconstituted in a hydrophilic interaction liquid chromatography (HILIC) solvent consisting of a 30:70 mixture of mobile phases A1 (10 mM ammonium acetate in 95% H_2O /3% acetonitrile/2% methanol + 0.2% acetic acid) and B1 (10 mM ammonium acetate in

93% acetonitrile/5% water/2% methanol + 0.2% acetic acid), supplemented with ^{13}C -lactate and ^{13}C -tyrosine internal standards. Reconstituted samples were centrifuged at 14,000 rpm for 5 minutes at 4°C and transferred to LC vials for mass spectrometry.

Targeted metabolomics data from bodies were analyzed using a 2×2 factorial design with treatment (light vs. dark) and genotype (ER2>Chr vs. ER2>WCS) as factors. Metabolites with more than two missing values across all samples were removed. Remaining abundances were natural log-transformed and mean-centered per sample. Residual missing values were imputed using KNN imputation ($k = 5$). A separate OLS linear regression ($y \sim \text{Treatment} \times \text{Genotype}$) was fit to each metabolite, with dark and ER2>WCS as reference levels. Benjamini-Hochberg FDR correction was applied independently to each coefficient family (treatment, genotype, and interaction). Metabolites with an interaction FDR < 0.05 were considered significant. Pathway enrichment was performed using the FELLA package in R, which leverages the topological structure of the KEGG metabolic network. The KEGG database for *Drosophila melanogaster* (organism code: dme) was compiled using buildGraphFromKEGGREST, excluding global overview pathways (map01100). Enrichment was conducted using the diffusion method, with the background defined as all metabolites in the input matrix with valid KEGG compound identifiers. Pathway nodes with a diffusion p-score < 0.10 were considered enriched.

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964 **Figures**

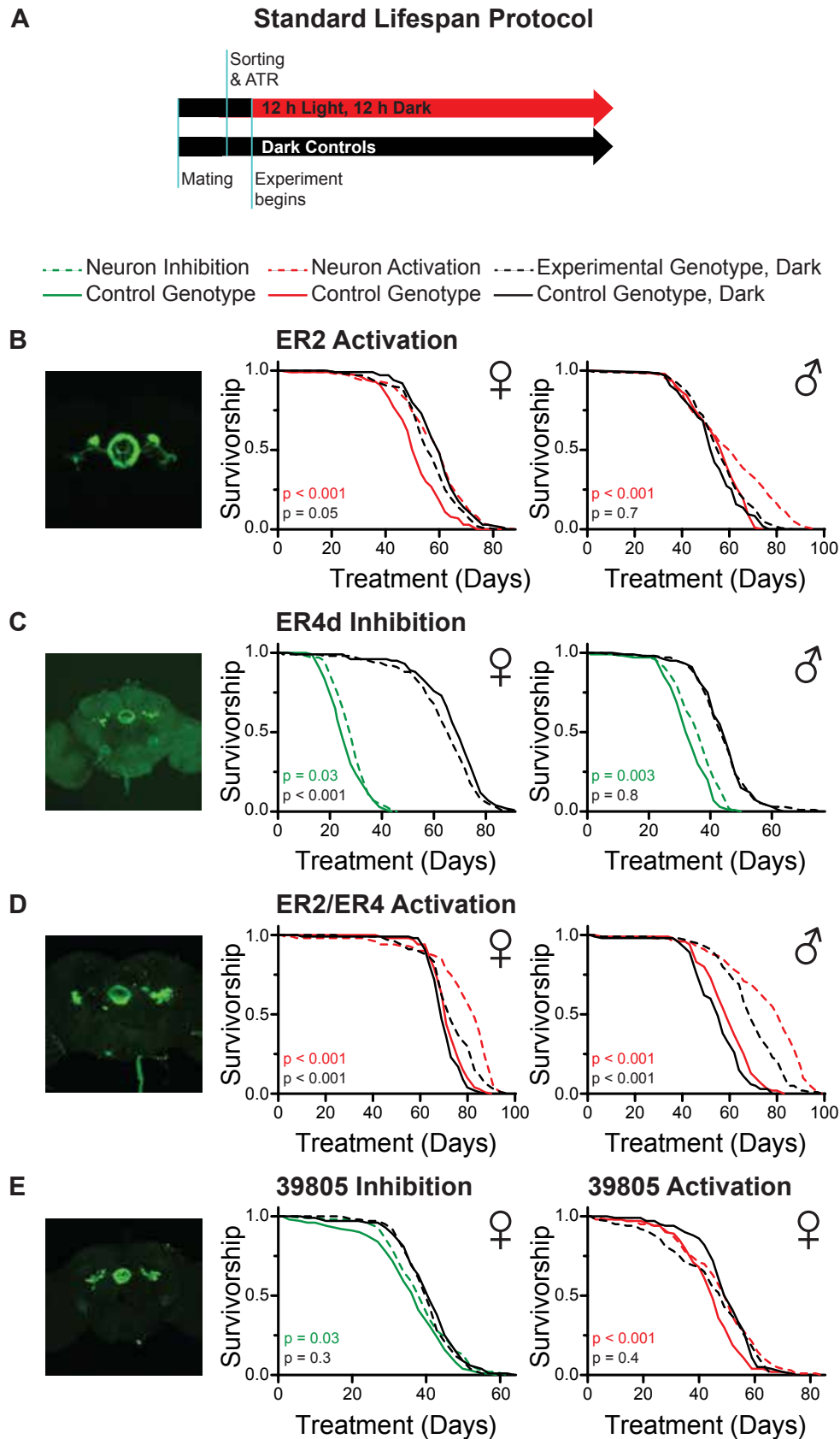


Figure 1. Manipulation of ellipsoid body neurons ER2 and ER4 extends lifespan in female and male flies. (A) Flies were mated for two days, then separated by sex and sorted 20/vial onto all-trans retinal food for 24 hours before optogenetic treatment began. (B–D) (Left) Expression patterns of drivers used in each experiment. Neurons were either activated (dashed red lines) using UAS-CsChrimson, or inhibited (dashed green lines) using UAS-GtACR1. Dark controls (black lines) were included for all experiments. P-values were obtained by log-rank analysis of correspondingly colored lines. Control genotypes (solid lines) were bred by crossing respective GAL4 drivers with WCS. (B) Activation of ER2 extended lifespan in both sexes. $n = 95-117$. (C) Inhibition of ER4d extended lifespan in both sexes. $n = 145-189$. (D) Simultaneous activation of ER2 and ER4 neurons extended lifespan in both sexes. $n = 127-139$. (E) Inhibition (left) and activation (right) of broader EB (BDSC #39805) modestly extended lifespan in females. GtACR1 $n = 127-133$; Chr $n = 95-115$. (B–E) P-values were obtained by log-rank analysis of line pairs indicated by color.

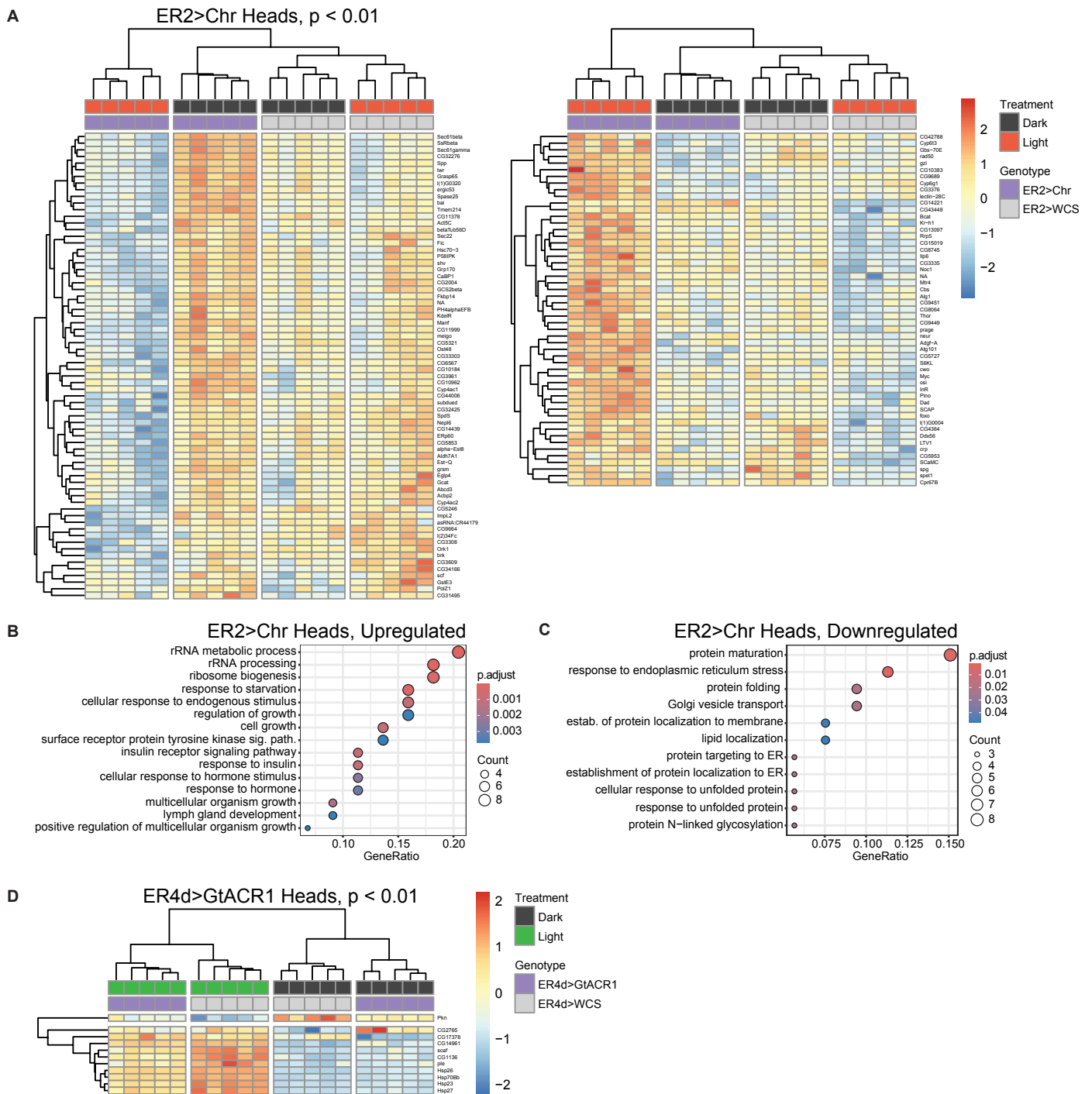


Figure 2. ER2 activation drives broad transcriptional changes and produces a head transcriptomic profile distinct from ER4d inhibition. Bulk RNA-seq data from whole heads of female flies treated with light for 1 week shows broad transcriptomic changes in ER2>Chr flies. Ten heads were collected per sample. Differentially expressed genes (DEGs) identified at FDR < 0.01 (Benjamini-Hochberg correction) using DESeq2 R package. **(A)** DEGs downregulated (left) and upregulated (right) by ER2 activation (ER2>Chr). **(B)** Top 15 GO Biological Processes upregulated by ER2 activation. **(C)** Top 11 GO Biological Processes downregulated by ER2 activation. **(D)** DEGs driven by ER4d inhibition (ER4d>GtACR1).

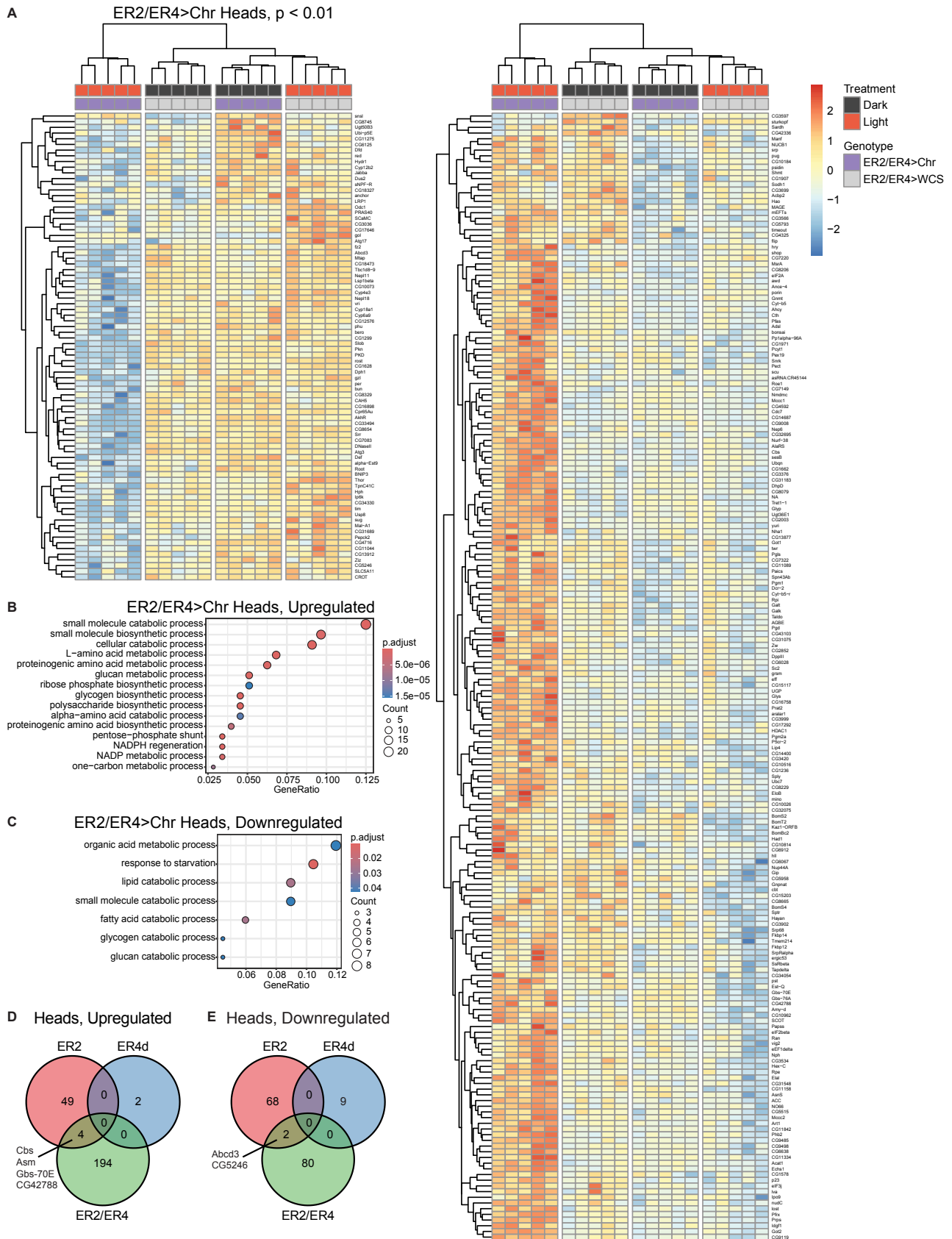


Figure 3. ER2/ER4 activation shares little transcriptomic overlap with ER2 activation or ER4d inhibition. Bulk RNA-seq data from whole heads of female flies treated with light for 1 week. Ten heads were collected per sample. Heatmap shows differentially expressed genes (DEGs) identified at FDR < 0.01 (Benjamini-Hochberg correction) using DESeq2 R package. **(A)** DEGs downregulated (left) and upregulated (right) by ER2/ER4 activation (ER2/ER4>Chr). **(B)** Top 15 GO Biological Processes upregulated by ER2/ER4 activation. **(C)** Top 7 GO Biological Processes downregulated by ER2/ER4 activation. **(D)** Overlap between upregulated head genes in all 3 manipulations. **(E)** Overlap between downregulated head genes in all 3 manipulations.

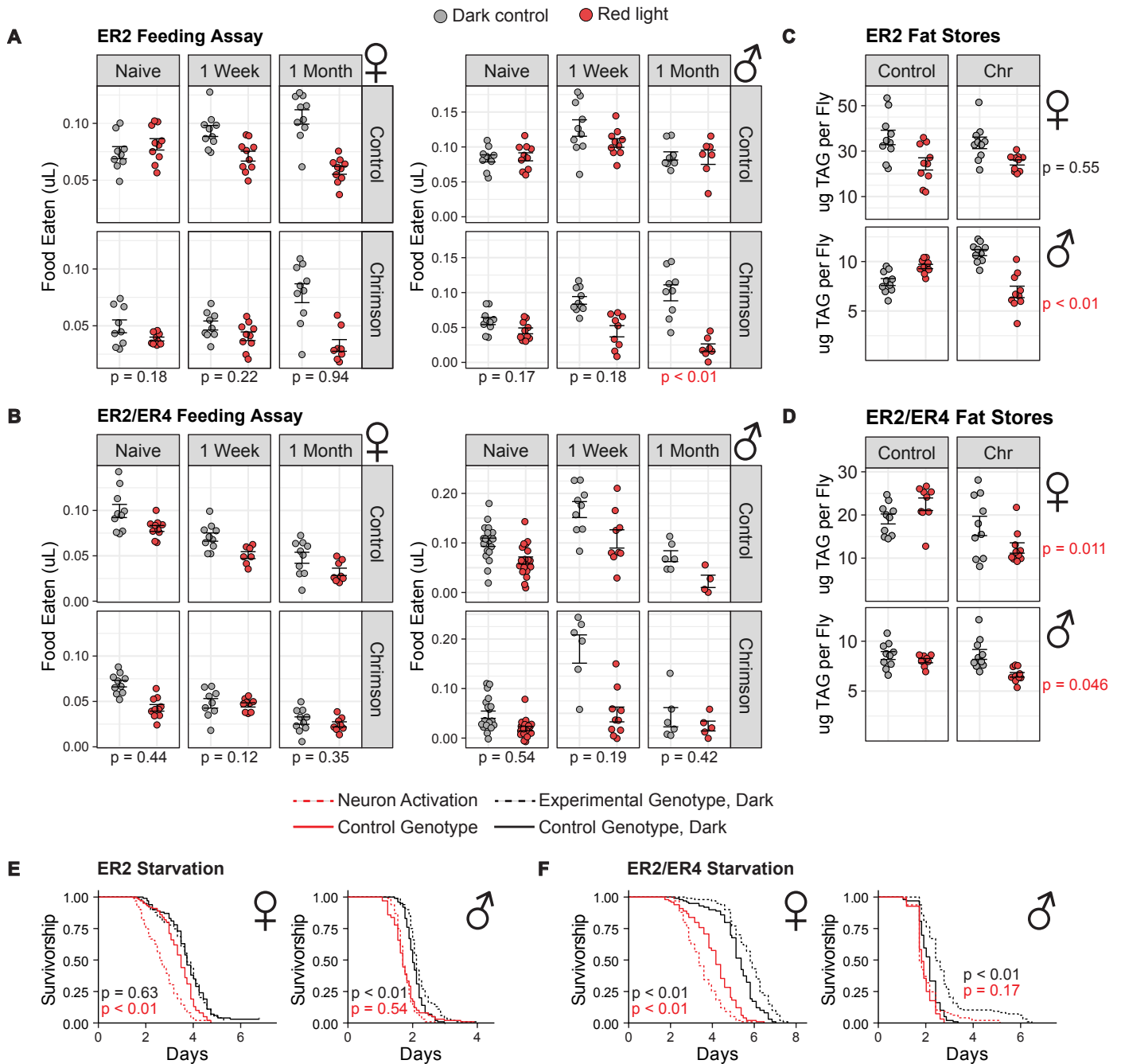


Figure 4. ER2 and ER2/ER4 activation dissociates nutrient-state signaling from behavioral and physiological outputs. (A-D) Interaction (light $\times$ genotype) p-values derived from 2-way ANOVAs. Benjamini-Hochberg correction was applied within each sex and driver, and across time points for con-ex. Female TAG points represent 5 flies; all others represent 10 flies. **(A)** ER2-GAL4 con-ex. ER2-activated females were indistinguishable from controls. Naïve and one-week pre-treated males were indistinguishable from controls, but male feeding was reduced after one month of pre-treatment. Female n = 8-10; male n = 7-10. **(B)** ER2/ER4-GAL4 con-ex. ER2/ER4 activated males and females were indistinguishable from controls at all time points. Female n = 7-10; male n = 18-20 (naïve), 6-10 (1 week), 4-6 (1 month). **(C)** ER2-GAL4 TAG levels were reduced for males, but not females. n = 10 for both sexes. **(D)** ER2/ER4-GAL4 TAG levels were reduced for both sexes. Female n = 9-10; male n = 8-10. **(E-F)** Starvation assays for ER2-GAL4 and ER2/ER4-GAL4 females and males. Flies were pre-treated with light for 1 week before transfer to starvation medium. P-values were obtained by log-rank analysis of correspondingly colored lines. n = 96-100.

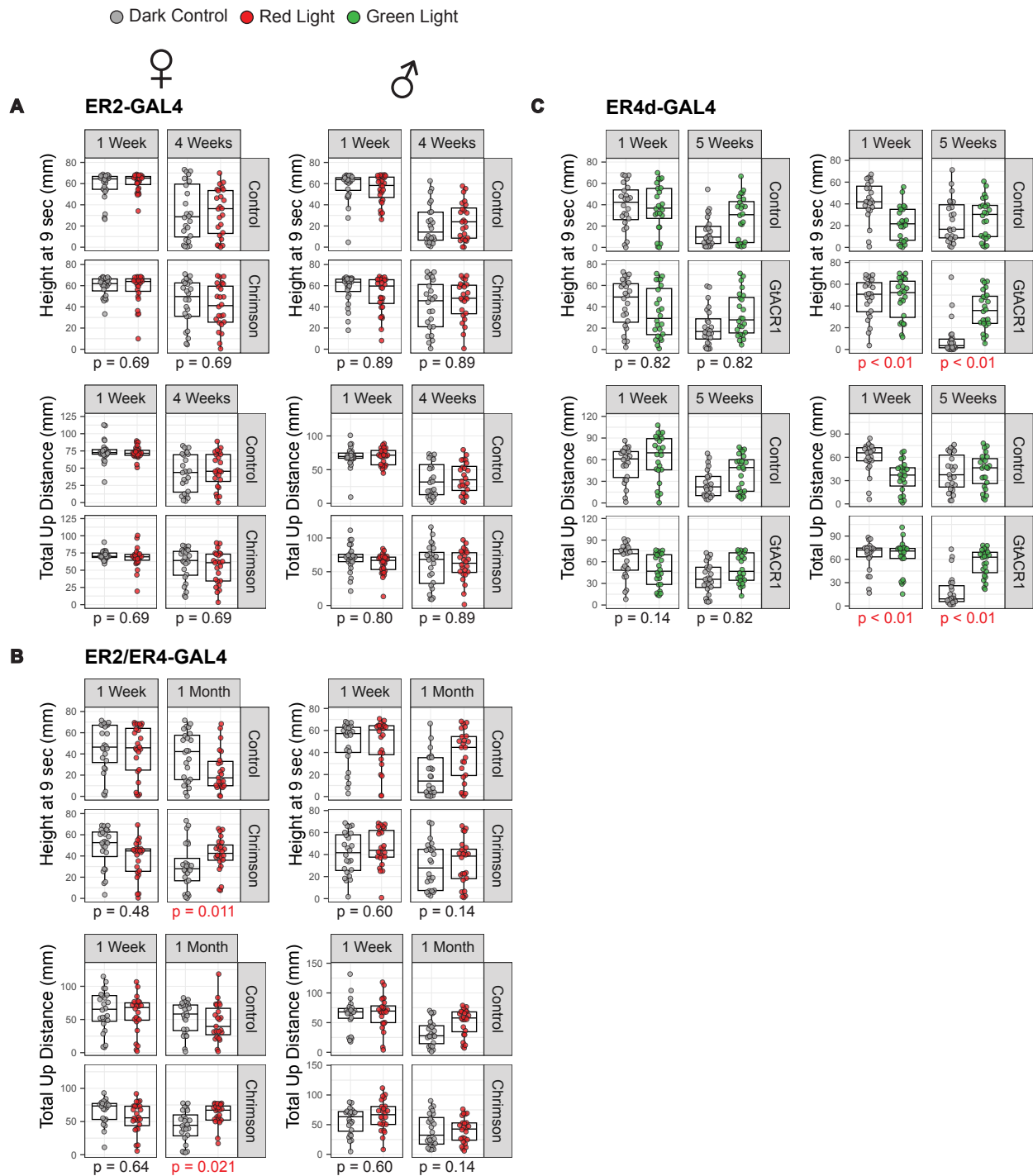


Figure 5. EB manipulation does not impair climbing ability. Height reached by 9 sec after drop (top plots of each panel) and total up distance climbed in 30 sec after drop (bottom plots). Interaction (light $\times$ genotype) p-values were derived from 2-way ANOVAs. Benjamini-Hochberg correction was applied within each sex/driver and across time points and climbing metrics. Points represent average of 3 trials per individual. $n = 23-28$. **(A-B)** One week of activation had no effect on ER2 and ER2/ER4 males and females. Female ER2/ER4 flies exhibit improved climbing by both metrics after one month of activation; ER2 females, ER2 males, and ER2/ER4 males exhibit no change after one month. **(C)** ER4d>GtACR1 males, but not females, show improved climbing ability after both one week and five weeks of inhibition.

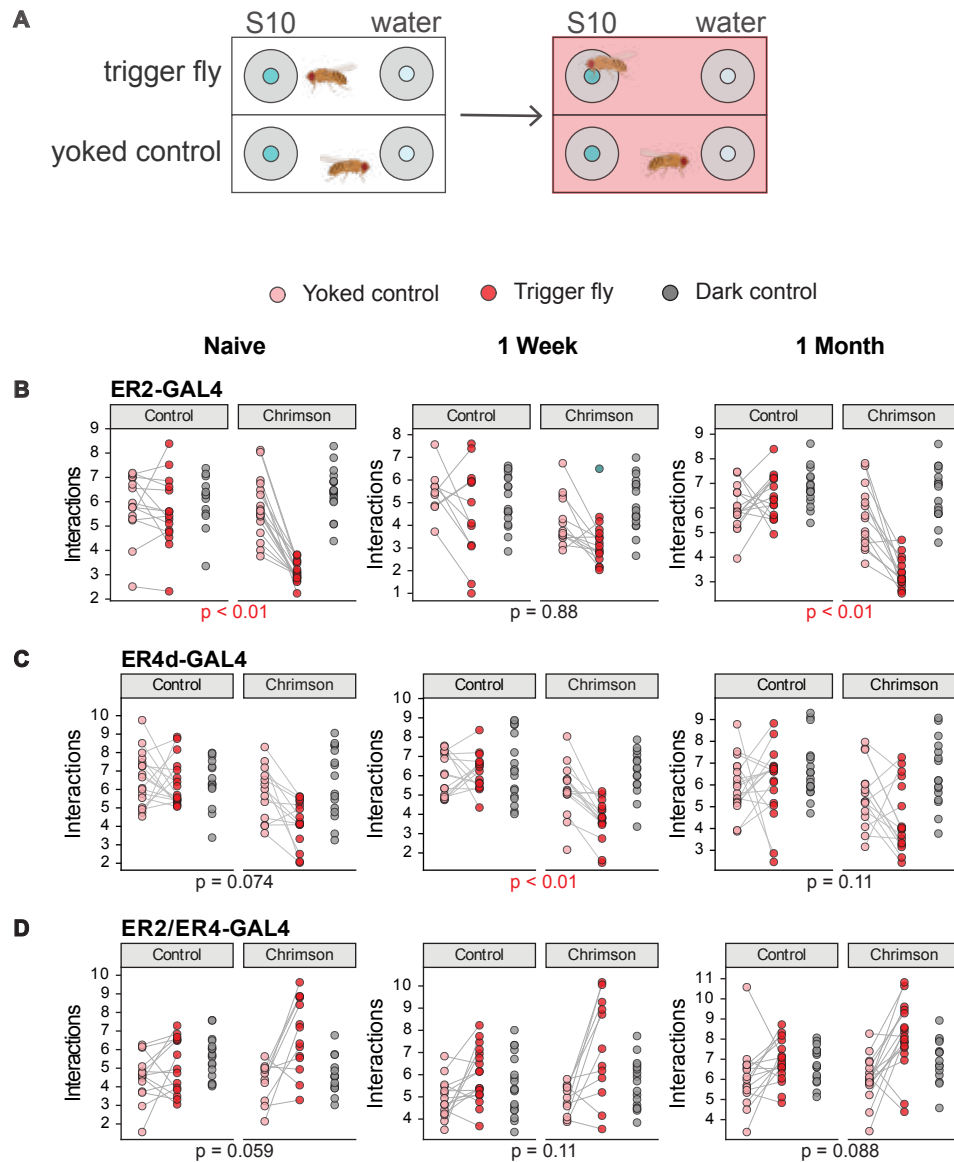


Figure 6. FLIC captures distinct valence states induced by EB manipulation. Female flies were used to assay valence of neuron activation. Sucrose interaction p-values were derived from the genotype $\times$ linkage interaction term of a linear mixed model (Interactions ~ Genotype * Linkage + (1|Group), lmer) with yoked pairs modeled as a random intercept. Benjamini-Hochberg correction was applied within each sex and driver and across time points. **(A)** A "trigger" fly (top chamber) controlled optogenetic light for itself and a yoked control (bottom) by touching a 10% sucrose (S10) well. **(B)** ER2 activation was aversive in light-naïve flies and in flies pre-treated with light for 1 month. $n = 9-18$. **(C)** ER4d activation was transiently aversive in flies pre-treated with light for 1 week. $n = 13-18$. **(D)** After correction, ER2/ER4 flies were indistinguishable from yoked controls. Before correction, light-naïve flies found the well rewarding ($p = 0.020$). $n = 10-18$.

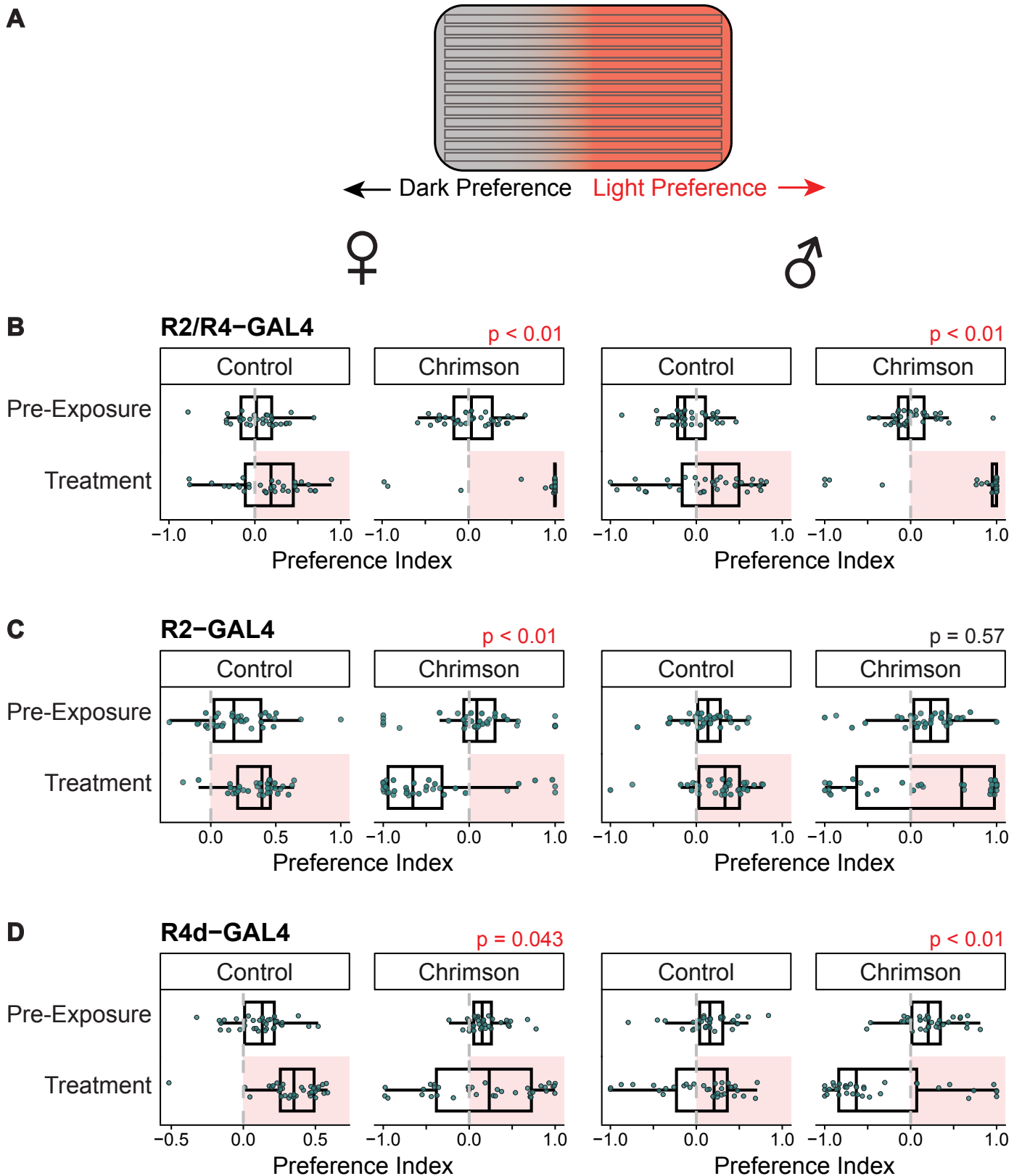


Figure 7. Valence of EB manipulation is sex-, circuit-, and context-dependent. (A) Light-naïve flies ($n = 34-36$) were placed in lanes with a dark side and an illuminated side (red light, 48 Hz) and subjected to a two-phase protocol: 1) a 10-min dark acclimation period; 2) a 60-min treatment of 5 sec on/5 sec off. Reported p-values for Preference Index correspond to genotype $\times$ phase interaction term. (B) ER2/ER4>Chr females and males strongly preferred lit side of arena. (C) ER2>Chr females, but not males, preferred the unlit side. (D) ER4d>Chr females preferred lit side, but showed decreased light preference relative to controls. Males preferred unlit side.

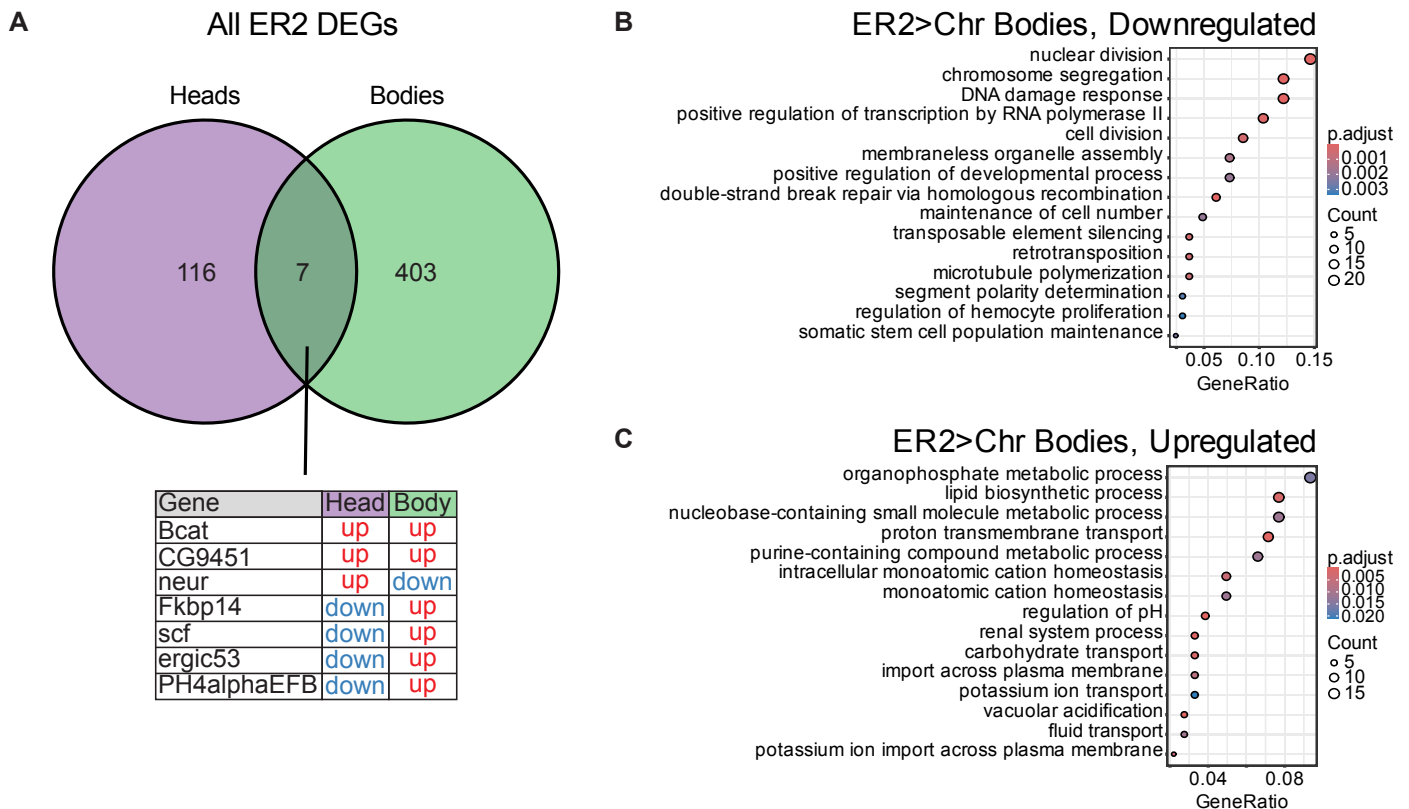


Figure 8. ER2 activation produces transcriptional signatures in the periphery that are distinct from those in the head. Bulk RNA-seq data were acquired from whole bodies of ER2-GAL4 flies treated with light for 1 week. 5 bodies collected per sample. Differentially expressed genes (DEGs) identified at FDR < 0.01 (Benjamini-Hochberg correction) using DESeq2 R package. **(A)** Overlap between all DEGs observed in ER2-activated heads (purple) and bodies (green). Shared genes are indicated by 2 red arrows (upregulated) or a red and blue arrow (oppositely regulated in heads and bodies; red = upregulated in heads, blue = downregulated in heads). **(B)** Top 15 peripheral GO Biological Processes downregulated by ER2 activation. **(C)** Top 15 peripheral GO Biological Processes upregulated by ER2 activation.

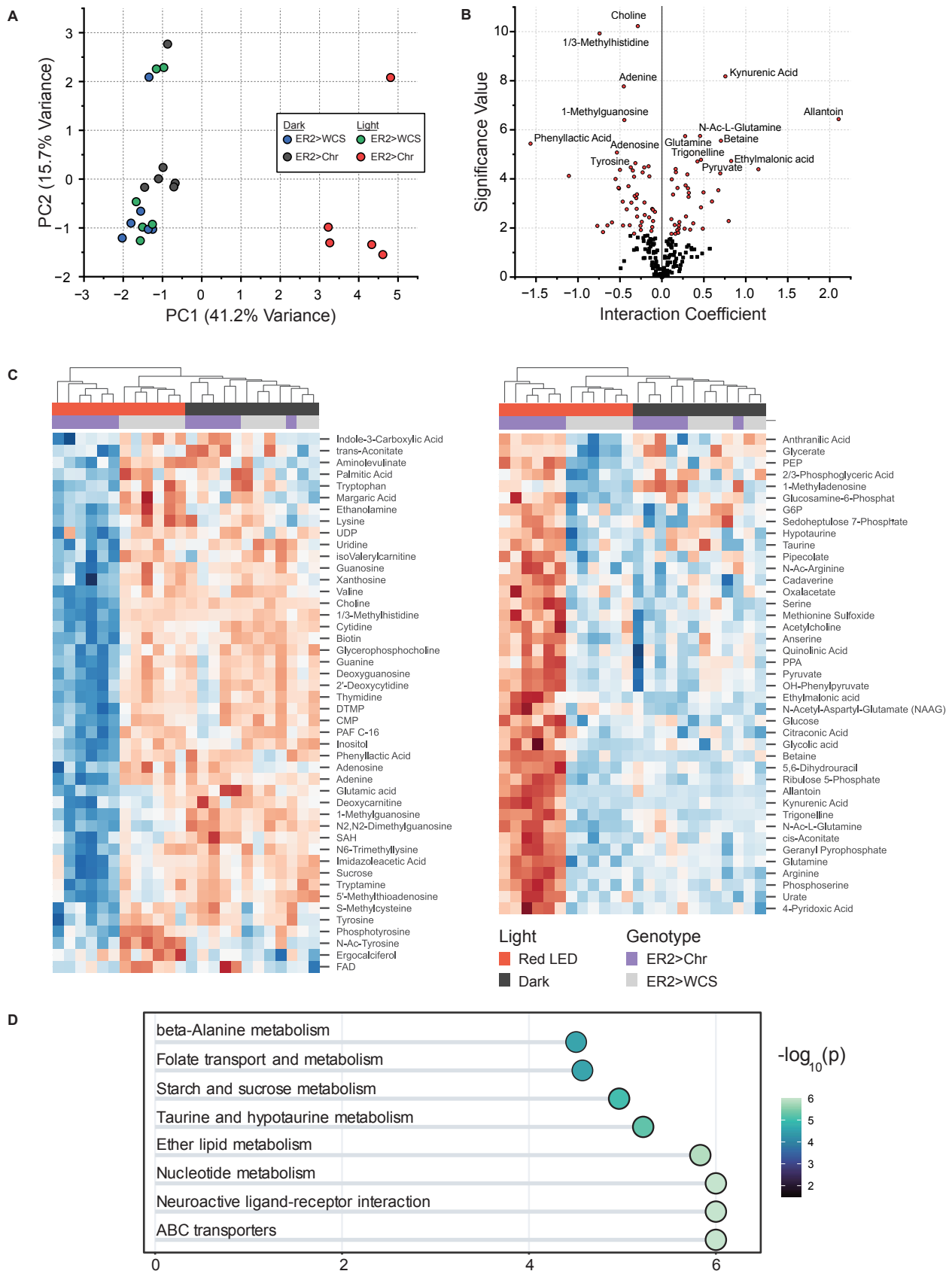


Figure 9. ER2 activation remodels peripheral metabolism. Targeted metabolomic profiling at FDR < 0.05 (light $\times$ genotype interaction; Benjamini–Hochberg correction). 30 bodies collected per sample. **(A)** PC1 separates ER2-activated flies from all other samples. **(B)** 87 metabolites were significantly affected by ER2 neuron activation (40 up/47 down). **(C)** Peripheral metabolites reduced (left) or increased (right) in abundance upon ER2 activation. Red light treatment separates ER2>Chr flies from all 3 control conditions. **(D)** KEGG pathway enrichment of significantly altered metabolites (FELLA). Dot color and x-axis indicate statistical significance ($-\log_{10}(p)$).

965 Supplemental Figures

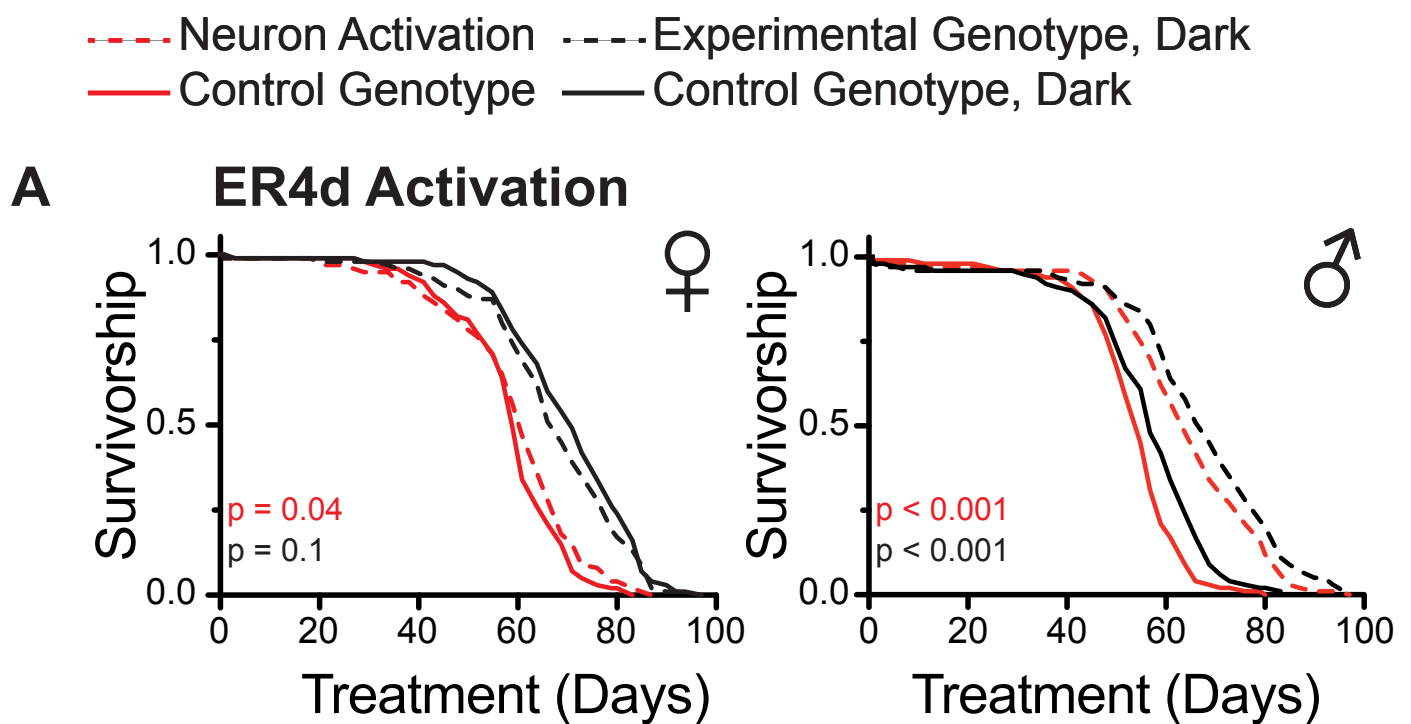


Figure S1. ER4d activation modestly extends lifespan in males and females. Flies were mated for two days, then separated by sex and sorted onto all-trans retinal food for 24 hours before optogenetic treatment began. Neurons were activated using UAS-CsChrimson (red lines). Dark controls (black lines) were included for all experiments. P-values were obtained by log-rank analysis of line pairs indicated by color. $n = 180-193$.

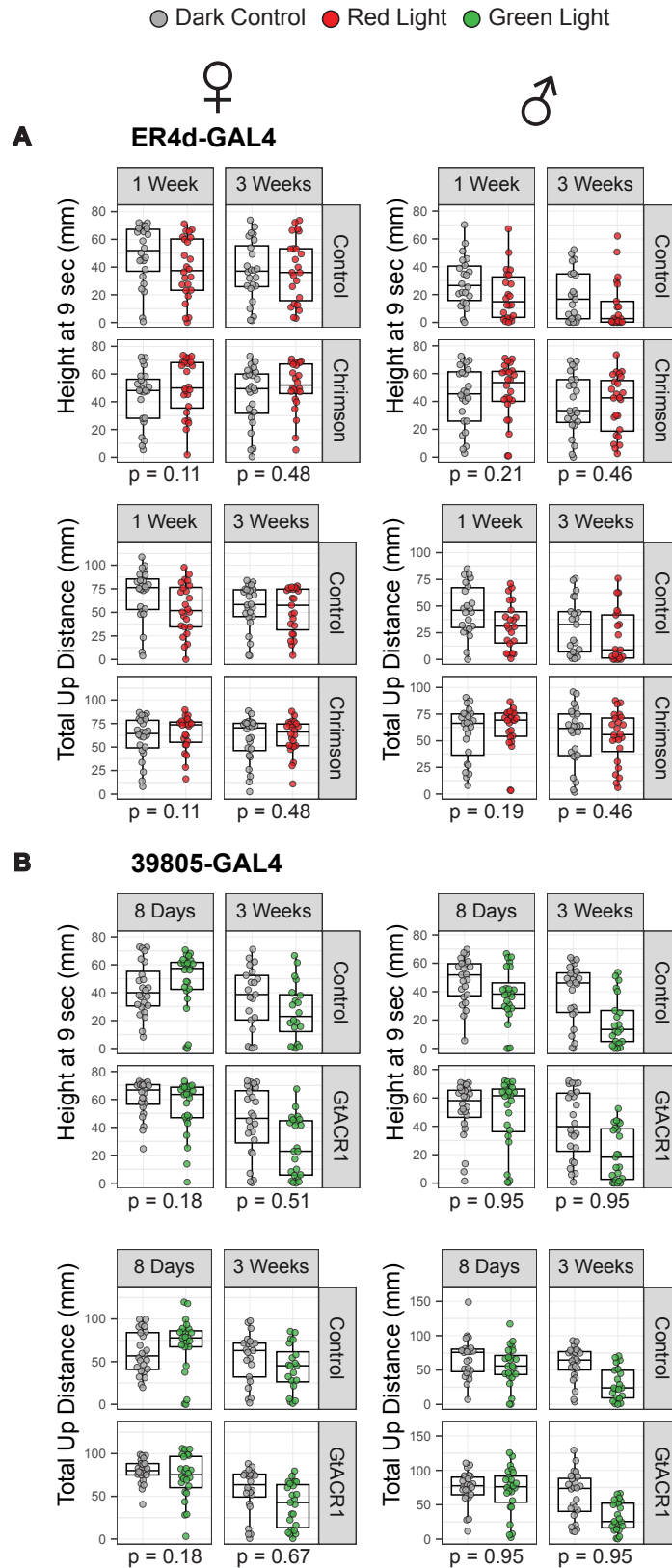


Figure S2. DDrop supplement. Height reached by 9 sec after drop (top plots of each panel) and total up distance climbed in 30 sec after drop (bottom plots). Interaction (light $\times$ genotype) p-values were derived from 2-way ANOVAs. Benjamini-Hochberg correction was applied within each sex and driver and across time points and climbing metrics. Points represent average of 3 trials per individual. $n = 20-24$. **(A)** ER4d activation has no effect on climbing ability of females or males. **(B)** Broader EB inhibition has no effect on climbing ability of females or males.

○ Yoked control ● Trigger fly ● Dark control

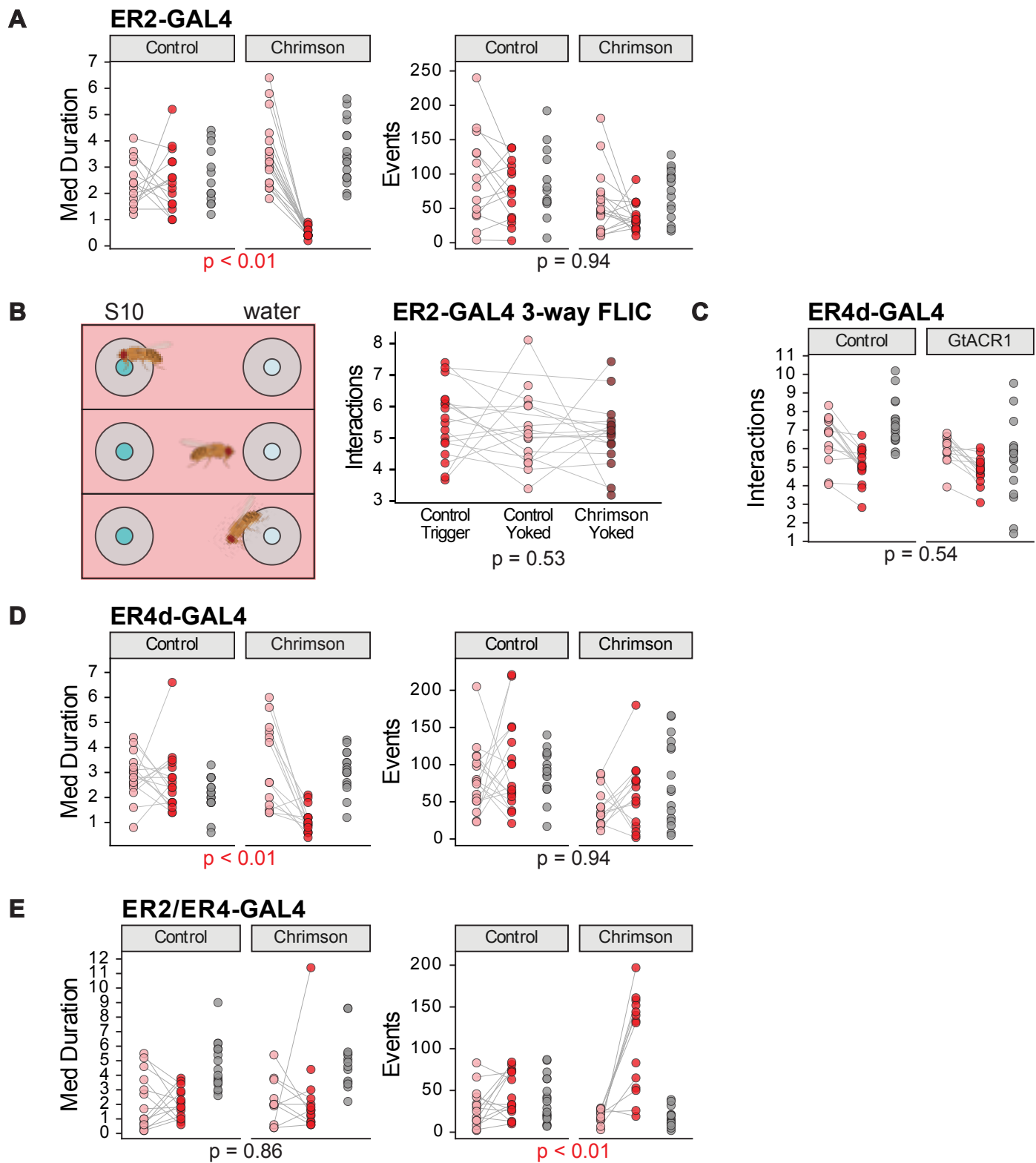


Figure S2. Self-administration of optogenetic activation in FLIC drives nuanced appetitive and aversive behaviors. (A, C-E) P-values were derived from the genotype $\times$ linkage interaction term of a linear mixed model (Outcome ~ Genotype * Linkage + (1|Group), lmer) with yoked pairs modeled as a random intercept. (A) Naïve ER2>Chr flies spent less time at the S10 well per visit (left), but approached it as often as controls (right). n = 13-18. (B) (Left) A "trigger" fly (top chamber, ER2>WCS) controlled optogenetic light for itself, a yoked genetic control (middle chamber), and a yoked ER2>Chr fly (bottom chamber) by touching a 10% sucrose (S10) well. (Right) Naïve ER2>Chr flies did not interact with the well more often than yoked genetic controls, indicating no global, non-associative effects of ER2 activation on feeding. P-values were derived from planned contrasts on a linear mixed model (Interactions ~ Condition + (1|Group), lmer) with the yoked pairing structure modeled as a random intercept. n = 16-18. (C) Naïve ER4d>GtACR1 interactions were indistinguishable from controls, as all flies were startled by onset of green light. n = 13-18. (D) Naïve ER4d>Chr trigger flies, like ER2 flies, spent less time at the well per visit (left) but returned as frequently as controls (right). n = 13-17. (E) ER2/ER4>Chr trigger flies spent the same amount of time per visit at wells (left) but visited more frequently than controls (right). n = 10-18.

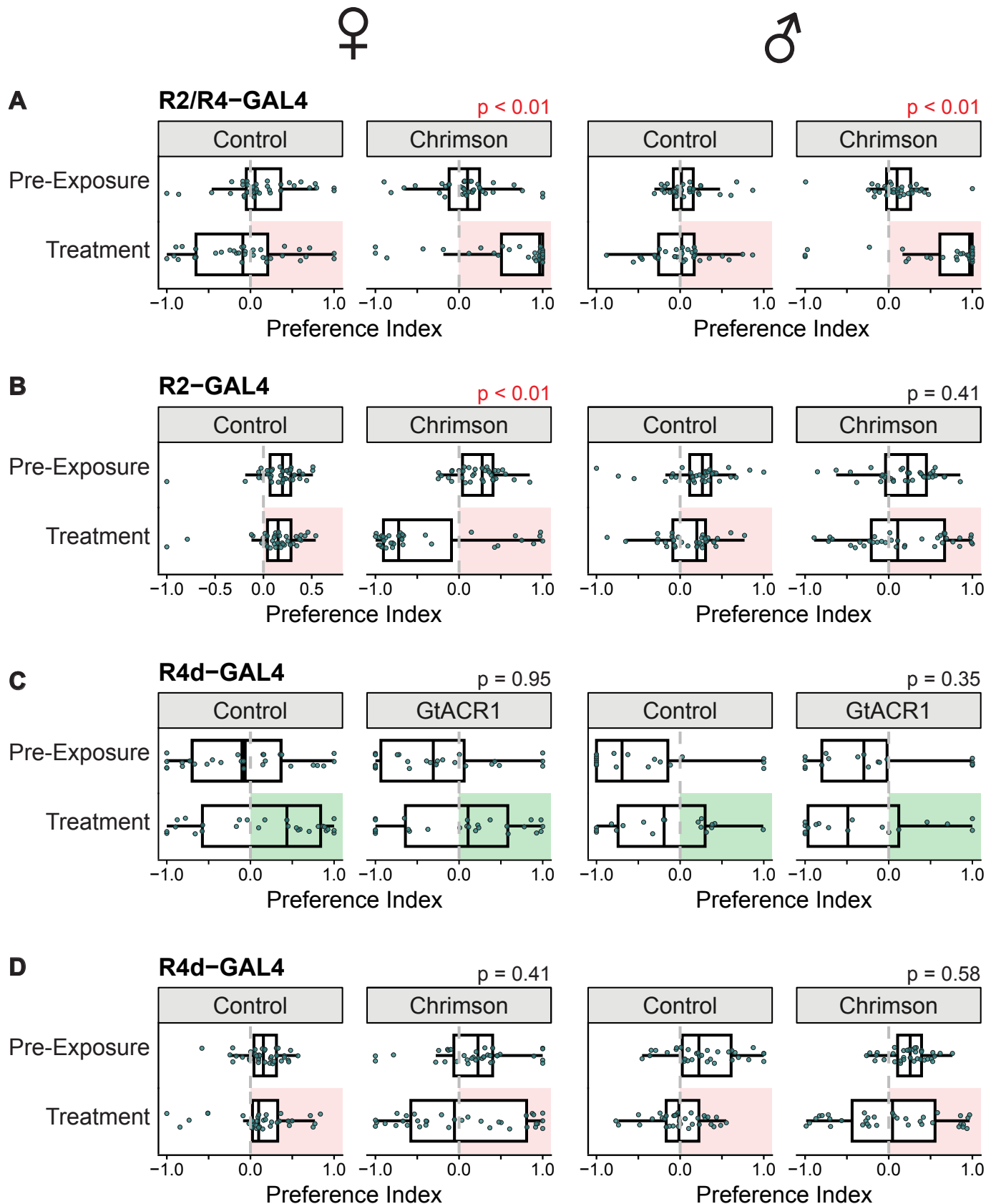


Figure S3. Stimulation frequency modulates EB-driven valence. Light-naïve flies were placed in lanes with a dark side and an illuminated side (red light at 200 Hz or green light at 40 Hz) and subjected to a two-phase protocol: 1) a 10-min dark acclimation period; 2) a 60-min treatment of 5 sec on/5 sec off. Reported p-values for Preference Index correspond to genotype $\times$ phase interaction term. **(A)** ER2/ER4>Chr females and males preferred the lit side. $n = 34-36$. **(B)** ER2>Chr females preferred the unlit side. Males were indistinguishable from controls. $n = 35-36$. **(C)** ER4d>GtACR1 females ($n = 26-27$) and males ($n = 17-18$) were indistinguishable from controls. **(D)** ER4d>Chr females and males were indistinguishable from controls. $n = 34-36$.

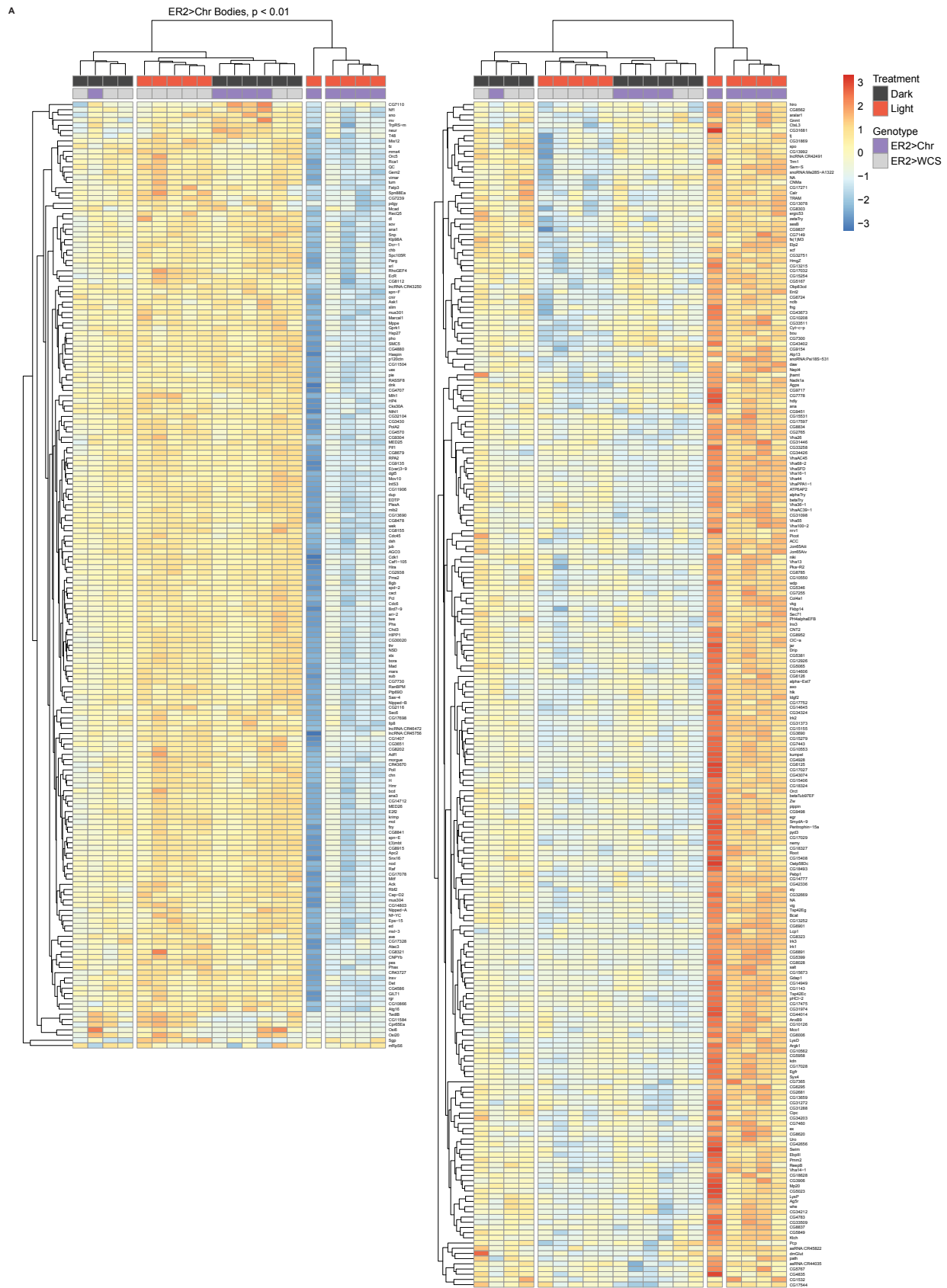


Figure S4. ER2 activation drives extensive changes in the peripheral transcriptome. Bulk RNA-seq data from headless bodies of female flies treated with light for 1 week. 5 bodies collected per sample. Differentially expressed genes (DEGs) identified at FDR < 0.01 (Benjamini-Hochberg correction) using DESeq2 R package. **(A)** DEGs downregulated (left) and upregulated (right) by ER2 activation (ER2>Chr).